

Review

Nature-inspired anti-fouling strategies for combating marine biofouling

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ABSTRACT

Marine biofouling is a major international challenge, which causes significant problems for global ecologies and economies. It manifests as the extensive and costly colonization of submerged surfaces with marine organisms, and it additionally results in the transportation of invasive, destructive, species around the globe. Effective, passive, antifouling treatments have been elusive - so far, only toxic treatments have been effective. To-date, long-term effective anti-fouling coatings are non-passive. These systems rely primarily on exfoliating surface layers (self-polishing) which continually release harmful metallic ions, along with associated booster biocides, into the local environment. Unfortunately, although these coatings can significantly reduce attached organisms on exposed surfaces, the resultant toxication of local marine environments and indiscriminate targeting of all proximate marine fauna, by these harmful systems, has depleted critical marine-based human food sources. The development of passive technologies is now beyond critical. Much of the ongoing research into the synthesis of alternative passive anti-fouling solutions has focused on naturally occurring, interfacial derivations. Evolutionary processes in nature have facilitated the development of broad-spectra, high-performance, and environmentally benign surfaces, and ongoing research is looking to adapt these novel structures. However, although this research has been extensive, an effective, long-term, passive technology continues to be elusive. This review provides insights into the anti-fouling processes, manufacturing techniques, and significant difficulties in the creation of the five nature-inspired anti-fouling systems. Several nature-inspired anti-fouling surface alterations are examined, along with evaluations of the drawbacks of each technique. The review concludes with insights into potential future lines of investigation for this globally critical field of study.

1. Introduction

Many marine organisms exhibit the sessile mode of life, which necessitates the availability of adherend surfaces, for which biota continually compete [1,2]. The “surface” in this context should also be understood to refer to any life form’s outermost layer, which must be kept clean in order for it to survive [3]. The resultant microfouling (the accumulation of bacteria, fungus, diatoms, and algae) and macrofouling (the accumulation of hydroids, barnacles, tubeworms, and macroalgae) can develop on sonar equipment, buoys, ships, and maritime

infrastructures, such as oceanographic sensors, and are a substantial global issue [4]. In addition, increased ship fuel consumption (increased drag) leads to higher carbon dioxide emissions and excessive maintenance costs. Furthermore, bio-invasion, the result of invasive marine organisms being transported around the globe, introduces aggressive, non-native fauna and flora into foreign, finely balanced, ecosystems. Anti-fouling (AF) paints have been created to combat marine biofouling [6–8]. The underlying operation is that the fouling organisms are toxically-eliminated before they can attach themselves permanently to the submerged surfaces [9].

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Biocidal agents prevent or restrict the settling of foulants by utilising chemically active substances. Although these compounds are quite effective against numerous organisms, they are non-specific and eliminate organisms that are not of their intended targets [10–12]. The widespread use of the broad-spectrum biocide tributyltin (TBT) since the 1960s to reduce marine biofouling is one of the most well-known examples. According to estimates, AF solutions save the marine industry \$60 billion in fuel and reduce 384 million tons of carbon dioxide emissions annually. In 2008, TBT was outright prohibited due to its significant toxicity to non-target marine species [13,14]. Restrictions on TBT have led to a resurgence of zinc and copper-based biocides in paints. The effectiveness against algae was increased by adding copper, zinc pyriithione, Irgarol 1051, and Sea-Nine 211 to the AF paints. Irgarol 1051 may have negative effect on marine plants with a lengthy half-life [15,16]. Treatment of wastewater tainted by AF chemicals in water, which costs worldwide maritime transport industry an additional € 3.5 billion [17]. Studies on biofouling were spurred by the harmful effects of toxic ingredients in conventional AF paints, and the need for “green” solutions.

In many situations, new AF techniques have been discovered through observation and research, such as the wings of cicadas and dragonflies, the leaves of lotus plants, the nepenthes inspired based slippery liquid infused surfaces (SLIS), cell membrane phospholipids, hydrogel like mucin and extracted biological macromolecule (EBM) [18,19]. Water droplets are spherical on a lotus leaf surface and they can easily roll off the surface because of the micro- and nano-architecture of lotus leaf, which reduces the droplet’s adhesion to the superhydrophobic surface [20]. A few insects, including dragonflies and cicadas, have developed nano protrusions on their wings that may burst and destroy microbes on touch without affecting their visibility [21,22]. SLIS are able to create an ultralow-fouling surface that effectively inhibits protein adsorption and bacterial adherence [23]. Cell membrane phospholipids inspired zwitterionic surface, which characterised by their neutral charge resulting from the presence of equal quantities of positive and negative charge on a single molecule and use in antifouling applications [24,25]. The setae on the gecko’s body assist it sticking to the wall. Many fish exude mucus through the scales, which protects them against a multitude of environmental factors [24,26–29]. EBM, such as medetomidine (Selektope) is becoming commercialized, that reduces the settlement of barnacles [30–37]. However, this is a continuously evolving area, and new summaries are perpetually in need. In this review, the fundamentals of biofouling and the shortcomings of conventional AF coatings are discussed, along with the advancement and novel concepts of recent developments in nature-inspired AF coatings. Emerging, nature-inspired, modified antifouling techniques are also addressed, as these have the potential to tackle the current challenges associated with a nature-inspired approach. The associated AF techniques, manufacturing processes, and functionalities are also addressed. The benefits and

drawbacks of the measures are examined, along with the difficulties and future directions of research on AF coatings and an evaluation of prospective nature-inspired surfaces.

2. Biofouling: formation and effects

2.1. Biofouling formation mechanisms

Marshall et al. pioneered the concept of biofouling, which was expanded upon by Fletcher, Characklis, and Flemming [38–41]. Biofouling involves communities of heterogeneous or homogeneous populations of one or more living microorganisms that are strongly bind to biotic and abiotic surfaces (Fig. 1) [42–44]. The process of biofouling may be categorised into distinct stages.

Biofouling is possible on any immersed surface, as the aqueous marine environment is home to over 4500 species of marine microbes, including proteins and other organic molecules (polysaccharides and glycoproteins). It is sufficient for even a small number of these to be adsorbed onto the surface (*conditioning*) via physical forces (electrostatic attraction, van der Waals, hydrogen bonding) which then act to provide the foundations for subsequent micro-/macro- organism attachment. Surface adsorption of modifies the physicochemical characteristics of the adherent compounds, driving adhesion of microbes, though initially the process is reversible. Under favorable conditions, however, microbial cells may convert from reversible (Biofouling Phase 1) to irreversible (Biofouling Phase 2) attachment [44]. Irreversible attachment usually develops within hours of interaction after the reversible process. Microorganisms attach to the surface and initiate cell-to-cell signalling to govern cell growth, replication, and extracellular polymeric substances (EPS) generation. EPS include the proteins, polysaccharides, glycolipids, glycoproteins, and DNA. Polysaccharides, lipids, and nucleic acids secreted by microorganisms which then help to trap more organisms and particles [43]. The abundance of nutrients and ease of adherence allow the development of localised biofouling microcolonies. The biofilm thickness develops over time, with the proliferation of microorganisms and the release of planktonic bacteria. Ultimately the tertiary colonizers, such as larger marine invertebrates, barnacles, tunicates, sponges, bryozoans, polychaetes, and macroalgae complete the settlement and the growth on the substrate [45].

2.2. Biofouling effects

Marine species accumulating on a ship hull have a major environmental and economic impact on the maritime industry. The maritime sector loses billions of dollars annually as a result of biofouling [46]. When an organism adheres to a ship’s hull (Fig. 2), there is an increase in the ship’s roughness, frictional resistance and drag. Even minor biofouling levels, which manifest as a thin mucous layer, are estimated

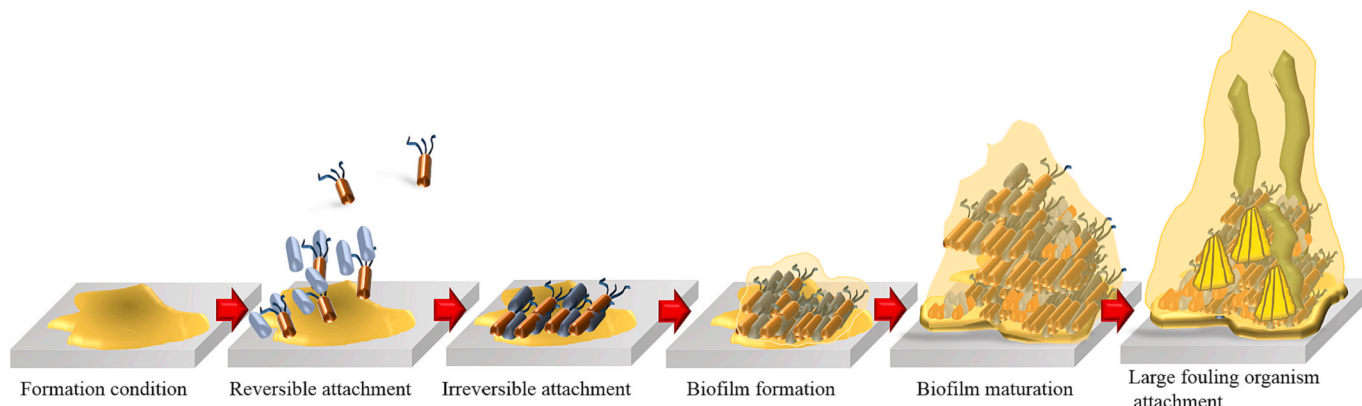


Fig. 1. Biofouling process in the marine environment.

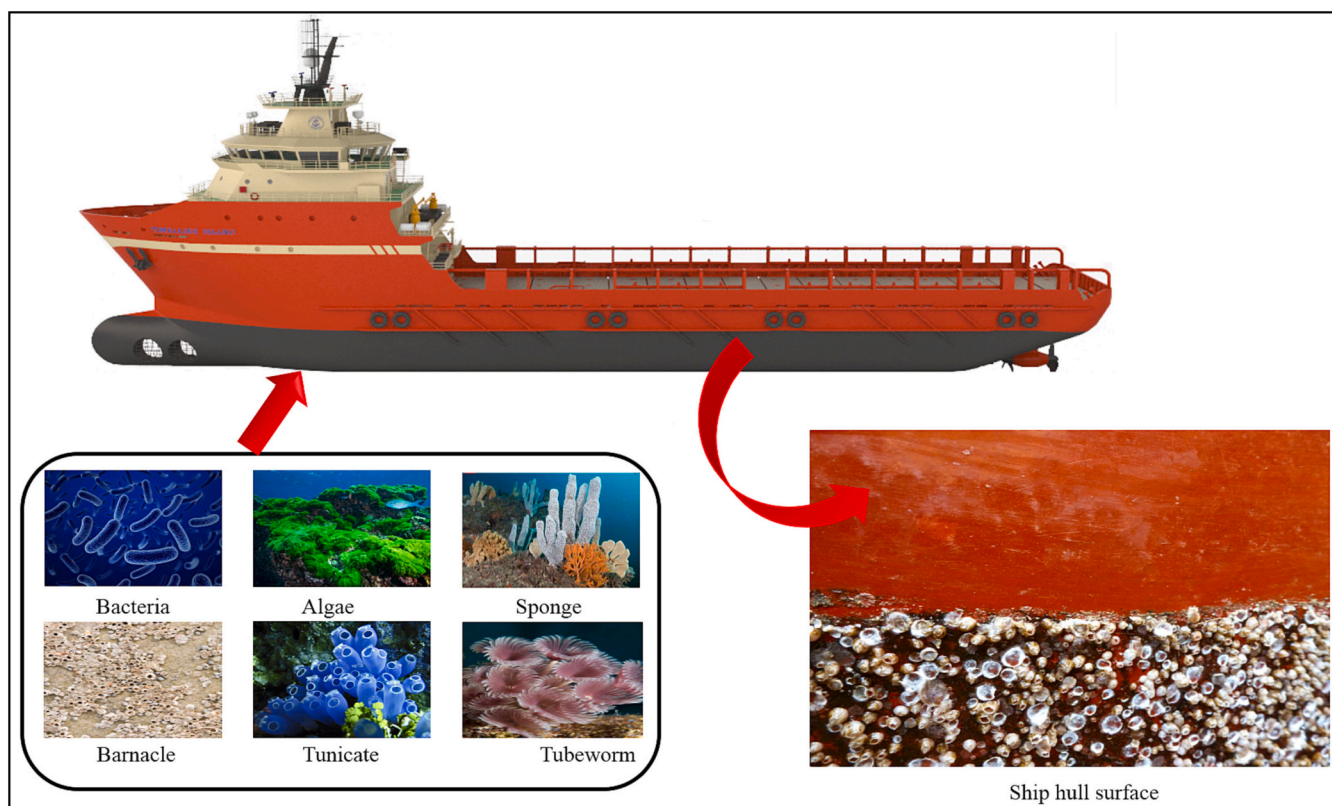


Fig. 2. (A) Schematic illustration of several marine fouling species adhering to ship hull surfaces and their harmful effects on the maritime industry [48]. Figure reproduced with permission from Springer Nature.

to increase the necessary shaft power by up to 11 % if to overcome the additional drag and maintain operating speed [17]. These consequences may result in significant financial losses owing to higher operational

costs, longer regular docking periods, as well as more human and mechanical effort [47].

Biofouling is a complex process that is governed by environmental

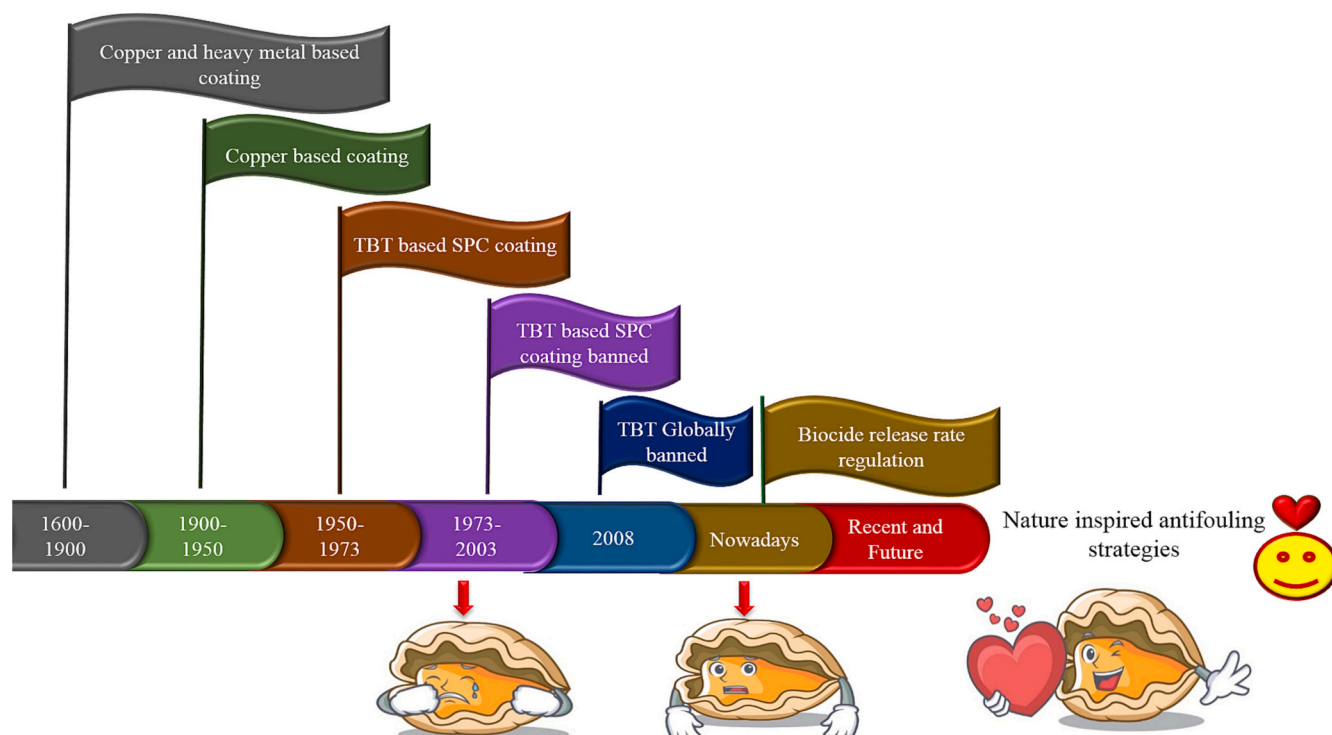


Fig. 3. History of AF strategies in marine.

conditions. The spatial and temporal diversity is a considerable challenge for the creation of innovative AF coatings, which must be capable of tackling diversified bio-fouling problems and have high environmental resilience. Preventative techniques and new therapy procedures must also be considered or improved.

3. Development of AF technologies

The evolution of AF technology is described in Fig. 3. Biocide-based techniques were recognised as early as the 7th BCE when ancient Mediterranean civilizations such as the Ancient Greeks and Phoenicians utilised the lead and copper sheets to reduce fouling (700–300 BCE). Seal tar, leather, and oil were employed to combat biofouling on wooden vessels between 1000 and 1500 (CE). In the late 18th and early 19th centuries, AF techniques using copper, arsenic, and mercury oxide were reported, but they had poor effectiveness, short shelf life, and high costs [49,50]. The first ship bottom paint was produced in 1908 arising from the successful development of spirit varnish paint. Numerous paint improvements were based on mercury oxide distributed in graded grain turpentine and alcohol. As a result, the lifespan of the paint was roughly nine months [51]. A US Navy study in 1926 showed that rosin was a cheap shellac alternative. For better performance, rosin was mixed with copper or mercury. Rosin was also used with synthetic petroleum-derived resins, which increased mechanical properties but also created health and safety concerns [9]. According to the discovery by Van Der Kerk and coworkers, the TBT organotin compound demonstrated a remarkable AF characteristic in the middle of the 20th century [16]. TBT was added to commercial AF paints in the late 1950s and was widely utilised in the middle of the 1960s to successfully reduce marine fouling on ship hulls for up to 5 years [52,53]. Oysters in Arcachon Bay (France) had growth issues, reproductive failure, and shell deformations in the late 1970s and early 1980s [54]. TBT generated by recreational boats in the neighbouring marinas was found to cause shell malformations and impede the survival and growth of *Crassostrea gigas* oyster larvae [55]. In addition, numerous investigations in the UK discovered a high correlation between TBT and the transformation of male sex organs to female ones (“imposex”) in marine gastropods (*Nucella lapillus*). This effect was ascribed to the interaction of TBT with molluscs’ endocrine systems [56,57]. After restricting the usage of TBT on small vessels, the quantities of oysters rebounded swiftly in France and the UK [58]. The International Maritime Organization (IMO) also found that TBT compounds could accumulate in humans and damage fish immune

systems [59]. The IMO assessed the negative impacts of TBT on maritime ecology in 2001. It recommended a restriction on the production of TBT AF paints from January 1, 2003, and a limitation on their usage on ship surfaces from January 1, 2008 [60,61].

Copper compounds have largely replaced TBT as the active agent, often in combination with algicides such as Irgarol 1051. However, the copper-based AF coatings were poisonous to the target species, although normal algae could adhere to them. Copper-based tin-free AF paints are scrutinised in several countries because of their greater release rates [62–64]. Therefore, the creation of novel, biocompatible, and ecologically benign AF coatings is greatly desired. Scientists have developed ecologically benign AF techniques in the past few decades, such as nature-inspired AF coatings [65,66]. Nature-inspired AF methods are gaining popularity, which are effective and eco-friendly. Novel materials may improve the AF activity of nature-inspired coatings.

4. Nature-inspired anti-fouling strategies

Both aquatic and terrestrial habitats have developed various strategies to resist colonization (Table 1). Inspired by natural processes, researchers have found novel AF solutions that are eco-friendlier and more sustainable in the marine environment (Fig. 4). This section focuses on the preparation of nature-inspired AF coatings and discuss their AF activities and challenges.

4.1. Nature-inspired slippery lubricant infused surfaces

(i) Inspired by nature

Slippery lubricant infused surfaces (SLIS) with low surface energy are textured surfaces that have been coated with a thin layer of lubricant [18,125]. In contrast to surfaces that reject liquids using re-entrant special-shaped microstructures, these slippery surfaces employ micro-textures to lock in lubricants [126–129]. The SLIS strategy is inspired by the renowned pitcher plant, such as the tropical carnivorous plant *nepenthes* (Fig. 5A–C) [130,131]. Its leaves are designed to attract, catch, keep, and digest arthropods. It uses the pitcher rim’s surface characteristics and insect “aqua planning”. To make the peristome super-hydrophilic, it requires a consistent architecture with smooth overlapping epidermal cells in the radial ridge. Rain may thoroughly wet this superhydrophilic surface under humid conditions to form a liquid layer that provides AF ability [132]. Normally, a slippery surface is

Table 1
Nature-inspired AF surfaces, as well as their origin and AF activities.

Nature-inspired AF surfaces	Origin	AF activities	Citation
Nature-inspired lubricate infused surface	Pitcher plant	<i>Escherichia coli</i> (<i>E. coli</i>), <i>Bacillus</i> spp., <i>Staphylococcus aureus</i> (<i>S. aureus</i>), Marine bacteria (<i>Marinomonas dokdonensis</i> (<i>M. dokdonensis</i>), <i>Cellulophaga lytica</i> , <i>Pseudoalteromonas</i> spp.), Barnacle cyprids (<i>Balanus improvisus</i>), Barnacles (<i>Amphibalanus amphitrite</i>), Diatom cells (<i>Navicula incerta</i> , <i>Nitzschia closterium forma minutissima</i> , <i>Phaeodactylum tricornutum</i> (<i>P. tricornutum</i>)), Zebra mussels, Mussels (<i>Geukensia demissa</i> , <i>Mytilus edulis</i>), Brown algae (<i>Cladophora</i> sp.), Red algae (<i>Porphyridium</i> sp.), Green algae (<i>Chlamydomonas reinhardtii</i> , <i>Chlorella sorokiniana</i> , <i>Chlorella pyrenoidosa</i>)	[28,67–84]
Cell membrane-inspired surface	Cellular membrane	Fluorescein isothiocyanate-labelled bovine serum albumin (FITC-BSA), <i>Bacillus subtilis</i> (<i>B. subtilis</i>), <i>E. coli</i> , <i>S. aureus</i> , <i>Pseudomonas aeruginosa</i> (<i>P. aeruginosa</i>), Marine bacteria, Diatom cells (<i>Navicula</i> sp., <i>Navicula incerta</i> , <i>P. tricornutum</i> , <i>Nitzschia closterium</i>), <i>Chlorella</i> , Mussels	[85–91]
Nature-inspired hydrogel surfaces	Seaweed, frog, carp fish	FITC-BSA, <i>E. coli</i> , <i>S. aureus</i> , BSA, <i>Nitzschia closterium</i> , <i>Nitzschia closterium forma minutissima</i> , Seaweed, <i>Chlorella</i> , Red algae (<i>Porphyridium</i> sp.), Diatom cells (<i>Navicula</i> sp., <i>Chlorella vulgaris</i> (<i>C. vulgaris</i>), <i>P. tricornutum</i>), Barnacle (<i>Balanus albicostatus</i>)	[92–104]
Nature-inspired textures	Sharkskin, Cicada and dragonflies, lotus leaf	<i>Nitzschia closterium forma minutissima</i> , <i>P. aeruginosa</i> , <i>Pseudomonas fluorescens</i> , <i>pseudomonas</i> sp., <i>C. vulgaris</i> , <i>Balanus improvisus</i> , <i>Dunaliella tertiolecta</i> , <i>Navicula</i>	[105–113]
Extracted biological macromolecule	Derived from Marine invertebrate, Marine plant, marine bacteria derived and Terrestrial plant derived	<i>E. coli</i> , <i>S. aureus</i> , <i>P. aeruginosa</i> , ascidian <i>Ciona intestinalis</i> , bryozoan <i>Bugula</i> sp., Green algae (<i>Enteromorpha intestinalis</i>), Calcareous tubeworm (<i>Hydroides elegans</i>), ascidian <i>Botryllus</i> sp., Red algae (<i>Ceramium</i> sp.), <i>Nitzschia closterium</i> , <i>Anthothoe chilensis</i> , <i>C. vulgaris</i> , <i>Artemia salina</i> nauplii, <i>Flavobacterium</i> , <i>Mytilus edulis platensis</i>	[114–124]

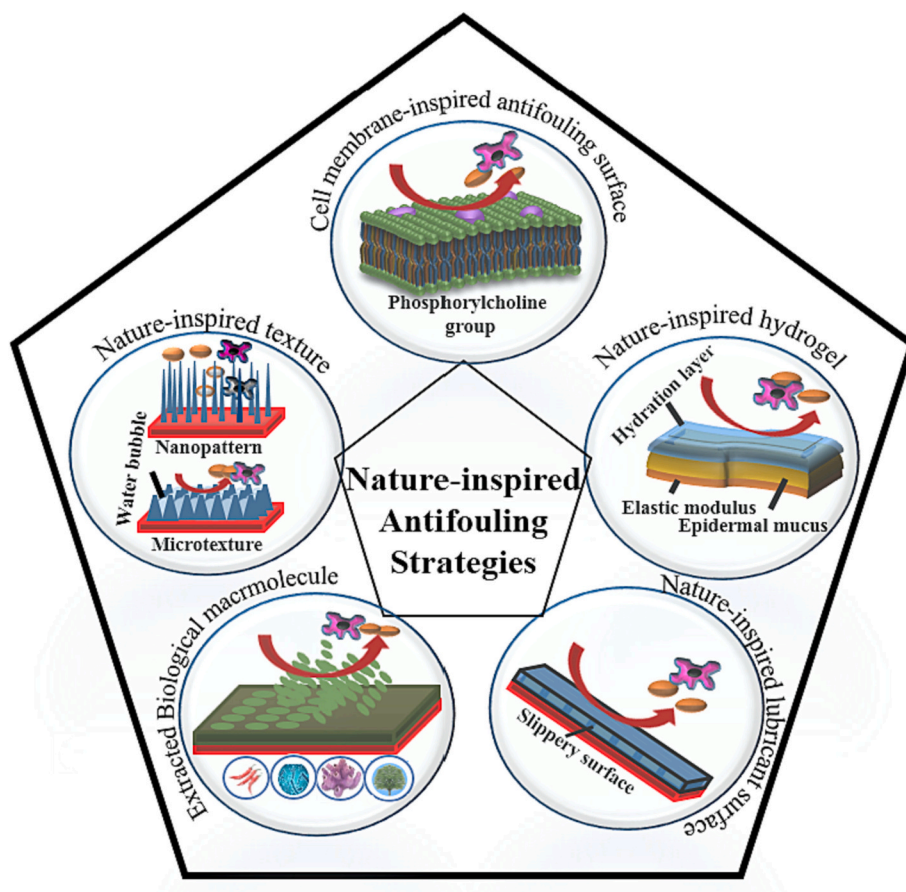


Fig. 4. The nature-inspired AF strategies for combating biofouling.

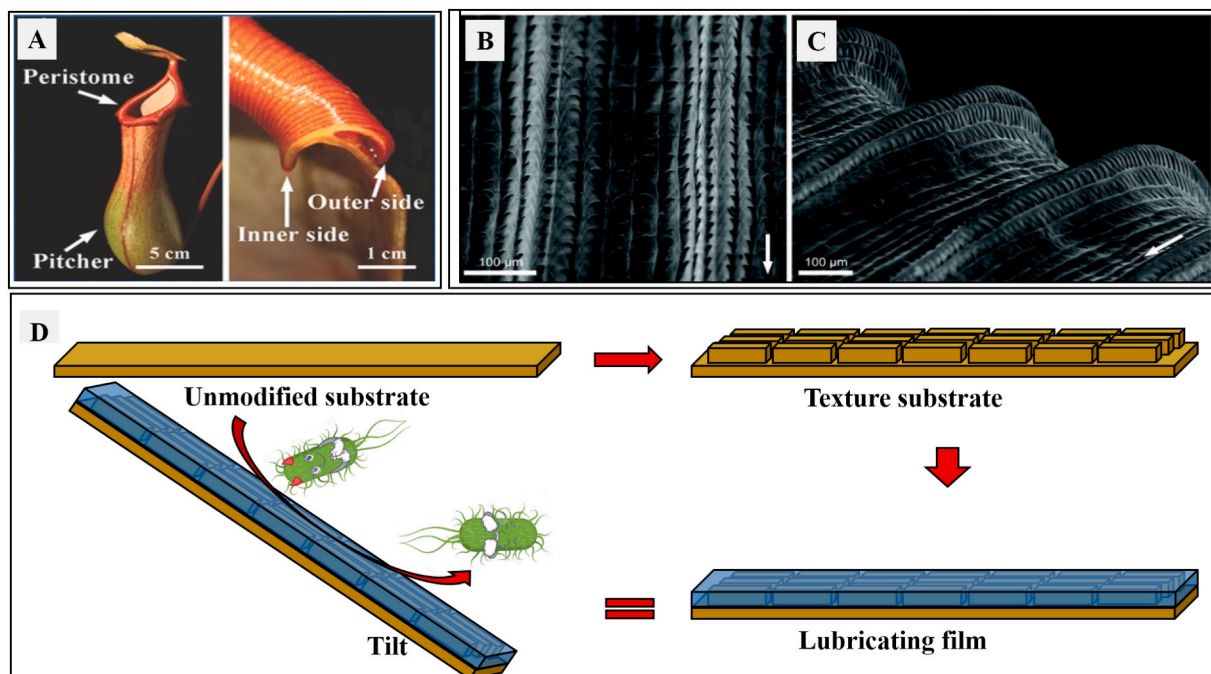


Fig. 5. (A) The pitcher of *Nepenthes alata* and the highlighted peristome surface [130]. Figure reproduced with permission from Wiley. (B–C) SEM analysis of the surface structures of the peristome. The direction of the pitcher plant's interior is shown by arrow [131]. Figure reproduced with permission from Springer Nature. (D) Generation of a SLIS on the polymeric surfaces to mimic the AF capabilities of the *Nepenthes* pitcher plant.

made up of a porous or textured substrate and lubricant (Fig. 5D). Fig. 5D depicts the process for producing SLIS, wherein a micro/nano-porous substrate is utilised to confine a lubricating liquid, resulting in the formation of a liquid layer [133]. As such, a brand-new group of surface that resists liquids and cleans itself has been developed. This lubricated surface is robust, defect-free, smooth, and chemically inert [134]. Furthermore, the lubricant layer may be repaired after physical damage cycles. This liquid-repellent surface may have benefited from fluid-handling instruments and fouling-resistant coatings under harsh conditions [135,136]. This section focuses on the preparation of SLIS, and their AF activities and challenges.

(ii) Preparation of SLIS

SLIS consists of two elements: substrate and lubricant. A super-hydrophobic substrate is first created, and is then infused with a lubricant. Surface free energy, roughness, and chemical composition homogeneity are all known to influence the wettability of super-hydrophobic surfaces [136,137]. Because of the presence of the surface liquid layer, the first two elements have a more pronounced influence on SLIS. If the surface is too rough, a suitable chemical coating may be applied to make it more suited for lubricant infusion [137–139]. There are several ways to create textured surfaces in the laboratory, including photolithography [140], plasma etching [141], chemical etching [142], and electrospinning [143], into which a lubricant is then infused to evolve the nature-inspired SLIS.

Metal substrates reported so far include carbon steel [144,145], stainless steel (SS), aluminium [146,147], titanium (Ti) [148–150], and magnesium [151]. To prevent fouling, marine-grade SS substrates were treated using impregnation of fluoro-silanization with a neutral per-fluorinated oil product [152]. Inorganic nonmetallic materials have also gained popularity in the maritime industry. Nonmetallic substrates such as polytetrafluoroethylene (PTFE), poly(methyl methacrylate) (PMMA), polydimethylsiloxane (PDMS), polycarbonate, and poly(vinyl chloride) have been continually explored for SLIS [72,153–157]. Vena et al. prepared a self-stratifying porous silicone (SPS) with a nonporous skin layer, using an emulsion template by allowing water droplets near the surface to evaporate to form the internal void structure. The liquid-infused system was then lubricated with silicone oil. Adjusting the surfactant and water concentration in the precursor emulsion can change the thickness, porosity, and pore size distribution of the nonporous skin layer of SPS. SPS has better lubricant locking ability and inject ability than nonporous PDMS to improve the overall system stability [158].

The stability of SLIS depends on the substrate qualities and surface micro-/nanostructures, as well as the physical and chemical properties of lubricants. The choice of lubricant is therefore critical in the preparation of SLIS, as a substrate-compatible lubricant may increase system stability [158]. Several lubricants including silicone oils, fluorocarbon oils and ionic liquids, have been utilised to construct the SLIS. Silicone oil is commonly utilised as a lubricant in SLIS owing to its outstanding qualities. However, it has a low surface tension, and quickly flows out of SLIS material throughout the working process, leading to lubricant loss, degradation of material property, and reduced service life. Furthermore, silicone oil cannot naturally decompose and be re-used, making them noxious to the environment [159]. Perfluorocarbons (PFCs) are excellent lubricants for SLIS materials. Among them, perfluoropolyether (PFPE) and perfluorodecalin (PFD) are popular. Large molecular weight PFPE demonstrate good viscosity-temperature property, thermal and chemical stability, broad temperature ranges, and low volatility. Although fluorocarbons have several excellent and distinctive qualities, they could chronically affect the environment and even humans. PFD could decompose thermally to perfluoro-butene and tetrafluoro-butene. Tetrafluoro-butene, a persistent carcinogen, hydrolyzes rapidly in water to form the fluorophosgene (COF_2) and reactive chemicals, which causes immediate lung damage (pulmonary edema) [160]. Ionic liquids are molten salts below 100 °C and made of bulky, asymmetric cations and

anions [161]. Ionic liquids have chemical inertness, high thermal stability, low volatility, and a broad operating temperature range (40–300 °C). Although ionic liquids possess exceptional qualities, their use as lubricants is mostly limited to cost-neutral applications and is not widely employed in conventional applications [162,163].

(iii) Anti-fouling activity

The slipperiness of the immobilised SLIS protect the surface against fouling organisms. LIS movement on the slick surface reduces fouling. By regulating the surface roughness, it is possible to diminish lubricant loss from the immersed SLIS under shear flow [164,165], confirming the long-term application of SLIS in marine environments. Epstein et al. [166] found that perfluorinated lubricant infused porous PTFE substrate could reduce bacterial adhesion. SLIS reduced *P. aeruginosa* biofilm by 99.6 % over seven days, under both static and physiologically relevant flow conditions compared to the uncoated porous PTFE (Fig. 6A–F). SLIS outperformed PEGylation-based therapies of *S. aureus* and *E. coli* by 1–2 orders of magnitude. They verified that the lubricant's remarkable mobility on the slippery surface reduced biofouling attachment. The slippery surface was durable under severe pH, salinity, and UV exposure. In another study, Zhao et al. [167] created microsphere surfaces from the poly(styrene-*b*-(ethylene-butylene)-*b*-styrene) elastomer and infused fluorocarbon lubricants into their porous structure to inhibit *E. coli* and *Chlorella ellipsoidea* adhesion for up to 10 days. The test results showed that the low-cost coating had remarkable AF capabilities. Furthermore, Wang et al. [147] used fluorocarbon as a lubricant and aluminium as a substrate to create a coating. This coating had a very low bacterial colony count after immersion in 200 mL 3-day-old culture medium for 3 and 7 days. The result suggested that the coating had successfully prevented sulphate-reducing bacteria from adhering to and growing on its surface. PFPE oil-infused superhydrophobic surface on SS was described by Ouyang et al. [168]. The population of deposited marine bacteria on SLIS was 2.25×10^7 cells/cm², which was an order of magnitude lower than the population on bare SS. The settling diatom densities on bare SS and SLIS were 5.28×10^7 and 5.50×10^5 cells/cm², respectively.

SLIS with silicone oil as lubricants has several AF applications. Ri et al. [169] added silicone oil to the dodecanethiol vapor treated Ti surface to provide an AF coating (SLIS Ti). SLIS Ti effectively inhibited biofouling of green algae and diatoms. After 14 days of immersion, 1.57×10^{11} cells/cm² of diatoms and 1.60×10^{11} cells/cm² of green algae were found on the bare Ti surface. The corresponding cell densities were just 6.84×10^7 and 5.05×10^{11} cells/cm² for the SLIS Ti. The cell adhesion density data clearly showed that the AF coating was imparted a 4 orders of magnitude decrease in cell densities over that of the bare Ti surface. Yang et al. [28] fabricated the superhydrophobic composite via solution blending of epoxy resin, hydroxyl-terminated PDMS, and nano titanium dioxide particles, then silicone oil infused into it. The SLIS epoxy-nanocomposite coating inhibited biofouling formation of *Bacillus* sp. and *P. tricornutum* by 93.6 % and 90.8 %, respectively. Furthermore, three different materials were tested by Amini et al. [170]: (i) a silicone oil-infused PDMS polymer network (i-PDMS), (ii) pure PDMS, and (iii) a commercial AF coating Intersleek 900 (IS900). After 8 and 16 days' marine field tests in Scituate Harbor, Massachusetts, USA, and these coatings were examined (Fig. 6G). SLIS had promise maritime AF applications, since i-PDMS showed the strongest antifouling behavior than pure PDMS and IS900.

With the help of researchers from many sectors, the nature-inspired slippery surface might provide new potential to combat marine concerns, improving the research field and biodiversity.

(iv) Recent issues of SLIS and their solutions

AF SLIS has gained popularity in recent years, and studies on SLIS are expanding, although they tend to concentrate on the substrate, morphology, and applications of the liquid-infused systems rather than

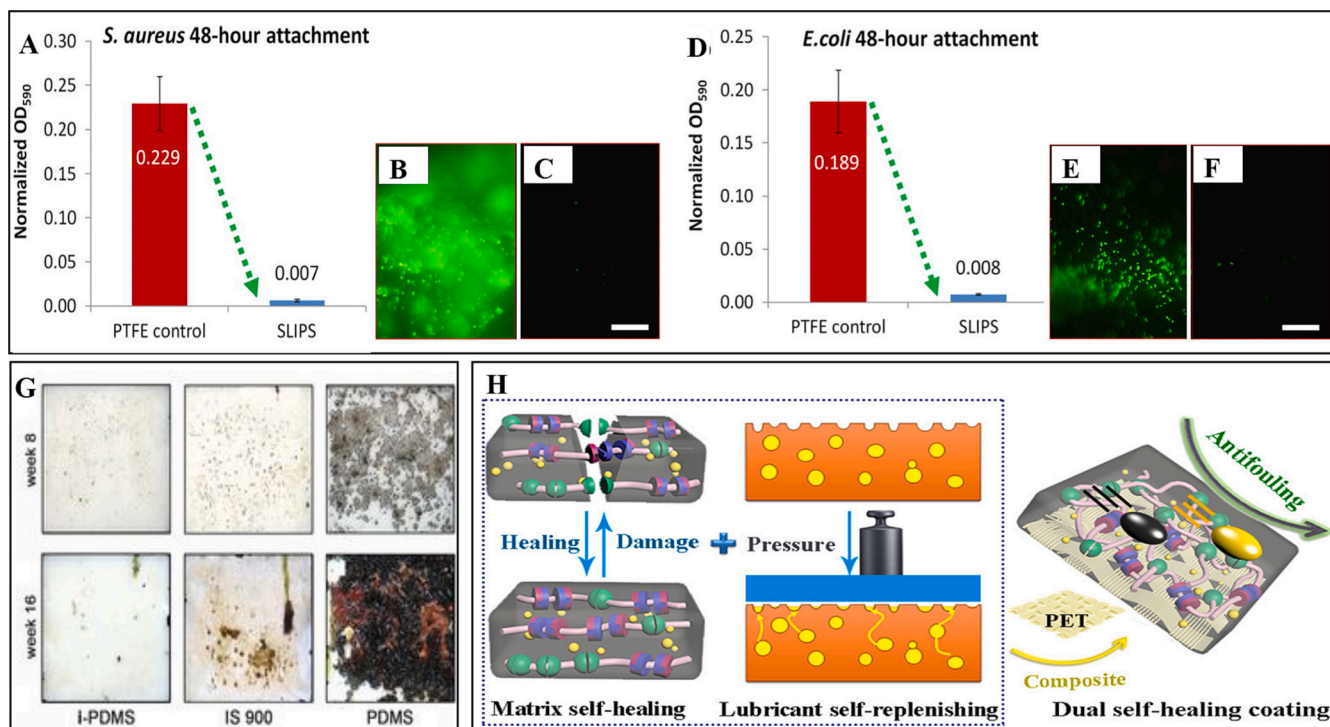


Fig. 6. Biofilm attachment of (A–C) *S. aureus* and (D–F) *E. coli* on PTFE and SLIPS; (B, C and E, F) Fluorescence images of attached *S. aureus* and *E. coli*, scale bar = 30 μm [166]. Figure reproduced with permission from American Chemical Society. (G) AF performance of i-PDMS, PDMS, and IS900 panels in marine field tests [65]. Figure reproduced with permission from Elsevier. (H) The self-healing properties of the matrix and the self-replenishing nature of the lubricant contribute to the slipperiness. The PDMS–IPDI–TPAL@Oil composite material is being investigated with antifouling activity [157]. Figure reproduced with permission from American Chemical Society.

the influence of lubricants during usage. Typical lubricant loss mechanisms include surface diffusion, external shear, gravity, and evaporation [159,171]. Its repellency depends on the lubricant, and its loss may result in SLIS failure. Marine SLIS must deal with high shear flows. After seven days, most lubricants lost their effectiveness in flowing water under shear stress, according to Keller et al. [172]. Under high-shear flows, lubricant depletion must be minimised [74]. The present study proposes two solutions to the lubricant loss problem. One is to boost the chemical connection and capillary effect between the lubricant and the substrate, to reduce lubricant loss. The second approach is to construct a porous surface that can hold fluids in the base material. When the surface lubricant is removed, the internal pores can retain fluid and replenish the top fluid layer via the linked porous architecture. Using silicone oil in thermoplastic elastomers, Yu et al. [157] created a slippery coating with lubricant self-replenishing and morphological self-healing capabilities. The polycondensation reaction of aminopropyl-terminated polydimethylsiloxane ($\text{NH}_2\text{-PDMS-NH}_2$) and isophorone diisocyanate (IPDI) in tetrahydrofuran solution containing silicone oil to produce polysiloxane–polyurea elastomer composite silicone oil (PDMS–IPDI@Oil). PDMS–IPDI@Oil with dynamic imine bond was produced by reacting the terminal -NH_2 groups with the -CHO groups of the chain extender terephthalic aldehyde (TPAL). Recombination of imine and hydrogen bonds, reversible cleavage, and rebuilding might repair the fracture (Fig. 6H). According to Fig. 6H, when the oil film was wiped off the surface with a paper towel, 500 g of pressure released the lubricant from the oil storage capsules, allowing quick replenishment of the lubricant layer to restore liquid repellence. It also demonstrates that PET@PIT@Oil has a good antifouling impact on proteins.

4.2. Cell membrane-inspired surfaces

(i) Inspired by nature

The cellular membrane is a highly intricate and nanostructured molecular complex found within living organisms. The composition of the structure primarily consists of phospholipid molecules, which serve a crucial function in segregating the intracellular cytoplasm from the extracellular milieu. Phospholipids possess hydrophobic alkyl chains and a hydrophilic polar group, and they undergo spontaneous assembly to form a continuous membrane when in an aqueous environment (Fig. 7A). Phosphorylcholine (PC) group is a significant polar group of phospholipids present on the cellular membrane. It is characterised by an electrically neutral and zwitterionic head group [173,174]. Zwitterionic groups have opposite charges molecules in their repeating units. Carboxybetaine (CB), sulfobetaine (SB) and PC having repeat units concurrently containing a quaternary ammonium moiety and carboxylate, sulfonate, or phosphate are the most commonly investigated betaines (Fig. 7B) [175,176]. These groups may attract water molecules, forming a thick hydrated layer on the material surface. The material became superhydrophilic to prevent adsorption of biomolecules and protein [177,178]. Zwitterionic-based materials are regarded as one of the most potent non-fouling materials due to their ease of synthesis and application. However, the high production cost of PC-based materials has limited their application. SB and CB compounds are less costly and their characteristics are unaffected by pH [173,179–181].

(ii) Synthesis of zwitterionic surfaces

Physical and chemical approaches are the most common strategies for using polymeric substrates in AF surface modification (Fig. 7C). The physical approach is often accomplished by depositing antifouling polymers on substrate surfaces via Layer-by-Layer (LbL) approach or spray, dip and spin coatings. LbL assembly for multi-layered thin coatings is gaining popularity due to its ease of preparation and ability to regulate surface thickness and composition.

The chemical approach involved surface anchored polymers via

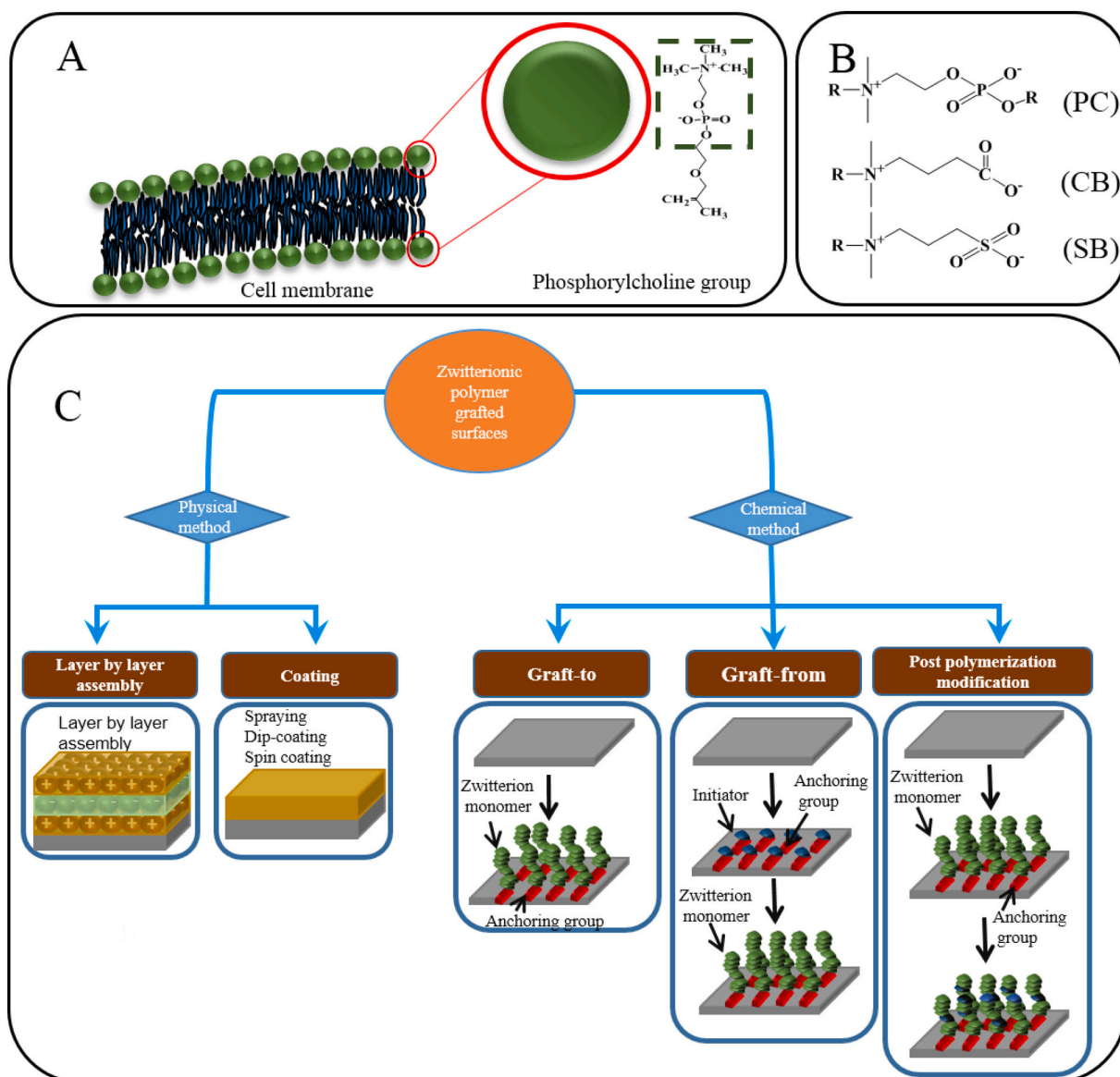


Fig. 7. (A) Schematic illustration of the structure of the cellular membrane lipid bilayer model, with PC groups on the outer and inner surfaces. The structure of PC group found in the cell membrane's outer leaflet. (B) Zwitterionic moieties and their chemical structures. (C) Surface grafting strategies.

grafting or specialised chemical processes, including “Grafting-to”, “Grafting-from” and post-polymerization modification. Xu et al. [182] constructed zwitterionic and cationic binary polymer brushes on polydopamine (PDA)-anchored SS substrates. The first synthesis of zwitterionic poly(MPC) included the polymerization process, which occurred by grafting from the surface of PDA-functionalised SS. This grafting was achieved by the use of a thiol-ene “click” reaction. The surface was further changed by grafting alkynyl-modified cationic poly(2-(methacryloyloxy) ethyl trimethylammonium chloride) (alkynyl-poly(META)) using an azide-alkyne “click” reaction. The PDA-modified surface was successfully grafted with poly(MPC)/poly(META) binary polymer brushes via click chemistry. The resulting polymer brushes exhibited remarkable stability, lasting for >30 days, and demonstrated strong resistance against fouling and bacterial growth. Dong et al. [183] used this method to create zwitterionic polymer surfaces on a variety of substrates for long-lasting anti-corrosion and anti-biofouling applications. A PDA layer was initially deposited on the substrates to increase anti-corrosion performance and function as a binder for attaching the subsequent layers. The substrates were then immobilised using 3-aminopropyltriethoxysilane (APTES) and poly(sulfobetaine methacrylate-co-

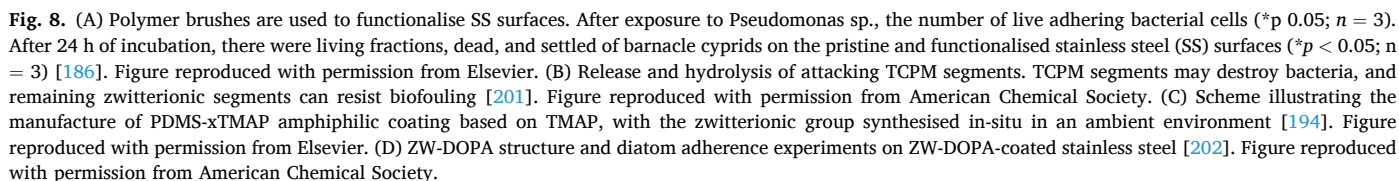
glycidyl methacrylate) (PSB) zwitterionic copolymer. In addition to utilising the “grafting-to” approach to incorporate the synthesised zwitterionic polymers into a matrix, the “grafting-from” technique was also employed to generate polymer brushes from the subsurface region of the matrix.

(iii) Anti-fouling activity

Biocontamination from proteins and bacteria affects biosensors, ships, and medical transplantation equipment. In real life, it is essential to combat biofouling and develop new, efficient anti-fouling materials. Most of the substances used to combat biological pollution are PEG and its derivatives [184]. PEG materials are unstable and easily oxidised in living environments. Zwitterionic polymers are an appropriate alternative for PEG. It has been discovered and confirmed that zwitterionic polymer materials have an anti-biological pollution mechanism. Zwitterionic polymers have numerous hydrophilic groups and strong dipole moments (such as sulfonic acid, amine, carboxyl, etc.). These groups may attract many water molecules, forming a thick hydrated layer on the material's surface. The material becomes superhydrophilic,

Zuilhof et al. [185] produced a self-healing AF zwitterionic polymer coating. They also reported a zwitterionic coating with anti-protein fouling and self-healing capabilities by depositing the polymeric colloidal particles on different surfaces. In another study, Wang et al. [186] produced the brushes by SI-ATRP of MPC and poly(ethylene glycol) methyl ether methacrylate (PEGMA) with varied feed crosslinker fractions. Subsequently, the crosslinked polymer brushes were successfully grafted onto the surfaces of stainless steel (SS). In tests conducted on barnacle cyprids and *Pseudomonas* sp., it was shown that the cross-linked polymer-coated SS surfaces exhibited superior efficiency compared to the non-cross-linked and uncoated pure SS surfaces (Fig. 8A). Under laboratory conditions, zwitterionic and neutral surfaces inhibited most of the foulants effectively. However, during the first week of field immersion experiments, anionic and cationic surfaces exhibited

superior anti-fouling characteristics. After eight weeks of immersion, all samples were completely contaminated, with no discernible difference [187]. Furthermore, Koc et al. [188] reported comparable findings with custom-made sulfobetain- and sulfobetaine-containing zwitterionic copolymer coating. The evaluation of a series of polyzwitterions incorporating sulfonate and sulphate groups demonstrated the significance of the polymer connection and/or spacer between the zwitterion and the polymer. It was observed that zwitterions containing a sulphate generally exhibited superior resistance to fouling compared to zwitterions containing a sulfonate under laboratory condition. However, when field experiments in the sea were conducted, the anti-fouling effect of the coatings failed [188,189]. More comprehensive analyses indicated that the accumulated foulants in field testing were mostly made up of diatomaceous soil and inorganic chemical sediments. Further laboratory simulations of the field experiments indicated that inorganic particulate



matters might collect on hydrophilic coatings and compromise the anti-fouling efficacy in <10 min. It is an exciting conclusion since the impacts of inorganic sediments are seldom explored in most biofilm or AF literature. On the other hand, polymers derived from PC, SB, or CB, have been extensively studied and exhibit excellent AF efficacy [67,186].

The need for elastomers in marine fouling-release coatings is substantial, because of their Young's moduli and low surface energies. In the field of maritime anti-biofouling coatings comprised of silicone elastomer-based fouling release agents (PDMS) are gaining popularity [153,190]. Deformation and flow shear rapidly eliminate adherent biofouling on silicone surfaces. However, it is challenging to the incompatibility of zwitterionic and silicone-based materials. The formation of a highly hydrophilic layer was possible by the using zwitterionic to change PDMS surface through the surface-initiated ATRP approach. Strong electrostatically-induced hydration was improved by zwitterionic chain-chain interactions and PDMS surface reconstruction was prevented [191]. For instance, Rahimi et al. [192] created another amphiphilic additive by connecting extremely incompatible blocks of hydrophobic PDMS and hydrophilic PSBMA utilising activators regenerated by electron transfer (ARGET) ATRP. The fouling-release performance of macroalgal spores (*U. linza*) and barnacles (*A. amphitrite*) was enhanced when an amphiphilic additive that self-migrated on the coated surface was combined with siloxane-polyurethane (SiPU) coating. Furthermore, it has been shown that the incorporation of the amphiphilic additive should take into consideration the molecular weight of the hydrophobic PDMS block. However, the anti-fouling performance of the material may not be influenced by the size of the hydrophilic PSBMA block. In another study, Rajan et al. [193] synthesised the PSBMA-*b*-PDMS-*b*-PSBMA triblock copolymer using ATRP, Michael addition, and ring open equilibrium polymerization (ROEP), and then integrated it into a polyurethane coating to test its stain-release ability. The modified polyurethane coating has a remarkable fouling-release effect on diatoms and bacteria, proving that movement of the PSBMA component on the surface affects the adhesion of marine foulants. In the future, it is imperative to make additional modifications to the PSBMA composition in forthcoming research endeavors, in order to achieve a coating that exhibits extensive cleaning capabilities. In 2020, Tan and colleagues synthesised hydrolyzable CB-ester ligands on SiO₂ particles, which were then incorporated into PDMS coatings. In-situ hydrolysis could create zwitterionic groups, which reduced mussel adhesion by 41.1 % after 14 days [89]. Furthermore, Wang et al. [194] developed a zwitterionic precursor, triisopropylsilyl 2-methyl-3-((3-(triethoxysilyl)propyl) amino) propanoate (TMAP), with a triethoxysilyl group and a triisopropylsilyl group on the CB ligand. The hydroxyl-terminated PDMS units could be polymerized with triethoxysilyl to form a uniform coating, and the triisopropylsilyl groups could be degraded into zwitterionic groups by self-catalytic processes. As a consequence, the coating containing 50 % TMAP had the greatest reductions in diatom and bacterium adherence (72.1 % and 72.0 %, respectively) (Fig. 8C). The absence of mussel colonization confirmed this coating's remarkable non-fouling property after two days of culture.

However, zwitterionic polymers, like other hydrophilic materials, have limited application in marine environments [67,186,195–198]. In reported marine field tests, *sea muck* and *Callysponginvaginalis* larva were found on the zwitterionic coatings after 2 and 6 weeks, respectively [91]. To address this issue, self-renewing and self-regenerating techniques for modifying the zwitterionic surface were developed [199,200]. In 2021, Dai et al. [201] created an anti-biofouling coating with fouling-releasing, fouling-killing, and fouling-resistant functions (Fig. 8B). A hyperbranched polymer that degraded in saltwater and AF groups to prevent and eliminate biofouling. Moreover, the liberation of biofouling on surfaces may occur throughout the process of deterioration, resulting in the regeneration of the surface. This coating exhibited great AF ability against common marine microorganisms and proteins when immersed in artificial seawater. Therefore, the creation of efficient zwitterionic coatings with long-lasting anti-fouling capabilities is

required for their useful purposes in hostile marine environment.

In hostile maritime settings, anti-fouling coatings should be designed for attachment strength. Mussel foot protein (Mfps) generates sticky plaques on maritime surfaces. 3,4-Dihydroxyphenylalanine (DOPA) enhances anchoring through semi-covalent coordination bonding and hydrogen bonding between substrates and catechol groups in Mfps. Yeon et al. [202] used a simple one-step procedure to create a zwitterionic dopamine derivative (ZW-DOPA) for constructing substrate-independent zwitterionic coatings via oxidative polymerization (Fig. 8D). They can be applied to a variety of substrates (e.g., nylon, titanium substrate, and SS). The addition of an oxidant (sodium periodate (NaIO₄) or (ammonium persulphate) modulate the oxidation of ZW-DOPA. The ZW-DOPA coating exhibited a significant level of resistance against the adherence of marine diatoms. After a 28-day exposure to saltwater, the coated SS surface exhibited a 52 % reduction in adherent diatom, in comparison to that seen on an uncoated substrate.

To summarise, super-hydrophilic coatings are often created using grafted hydrogels and brushes. Even though essential factors, including producibility, excellent adhesion on a variety of substrates, and long-term stability, have already been considered by researchers. More variables must be addressed in the future to enhance the long-term relevance of super-hydrophilic anti-fouling coatings in the maritime sector. The research on cell-membrane inspired (zwitterion) surfaces and its impact on biofouling development is summarised in Table 2.

4.3. Nature-inspired hydrogel surfaces

(i) Inspired by nature

Many creatures in nature have superwettability and unique adhesive surfaces. Many of them, including fishes [203], seaweed [204], and tree frogs [205] have nature-inspired hydrogel surfaces that have been studied to uncover the underlying processes (Fig. 9A). For example, under damp conditions the *Litoria caerulea* (tree frog) can glide on branch surfaces and flat leaves without falling, even though the toe pads are completely mucus-coated. The reversible moist adherence phenomenon might arise from the presence of regular micro/nanostructures and hydrogel-like mucus, which promote boundary friction and tight contacts between the surface and epidermis [205–207]. The seaweed *Saccharina japonica* (*S. japonica*) has salt-tolerant hydrogel surfaces [204]. It has a thin layer of hydrogel-like mucilage rich with polysaccharides (alginate, agar, and carrageenan). The presence of a mucilage layer facilitates the retention and interconnection of a significant quantity of water molecules, so imparting a liquid-like characteristic to their surfaces. Consequently, this phenomenon leads to the manifestation of innate submerged superoleophobicity. Carp scales exhibit superoleophobicity, due to their natural habitat, and are good candidates for investigating underwater superwettability. Due to its self-cleaning ability and superoleophobic skin, the carp is able to thrive in oil-polluted environment [203]. Carps have tightly organised fan-shaped scales of 4–5 mm in diameters, on which are orientated micro/nanotexture papillae with widths of 30–40 µm and lengths of 100–300 µm. The scales are typically coated with a thin layer of hydrogel-like mucus. A liquid-like surface with superoleophobicity in water and

Table 2
Examples of cell-membrane inspired antifouling surfaces.

Zwitterionic compound	Antifouling activity	Citation
Sulfobetaine methacrylate-co-3-sulfopropyl methacrylate	Red algae <i>Porphyridium</i> sp.	[91]
Modified-SiO ₂ (N-methoxyacylethyl)-3-amonopropyltriethoxysilane, Polydimethylsiloxane	FITC-BSA, <i>Pseudoalteromonas marina</i>	[89]
Polydopamine/poly(sulfobetaine methacrylate)	BSA, <i>Pseudomonas aeruginosa</i> , <i>Bacillus subtilis</i> and <i>Navicula</i> sp	[87]

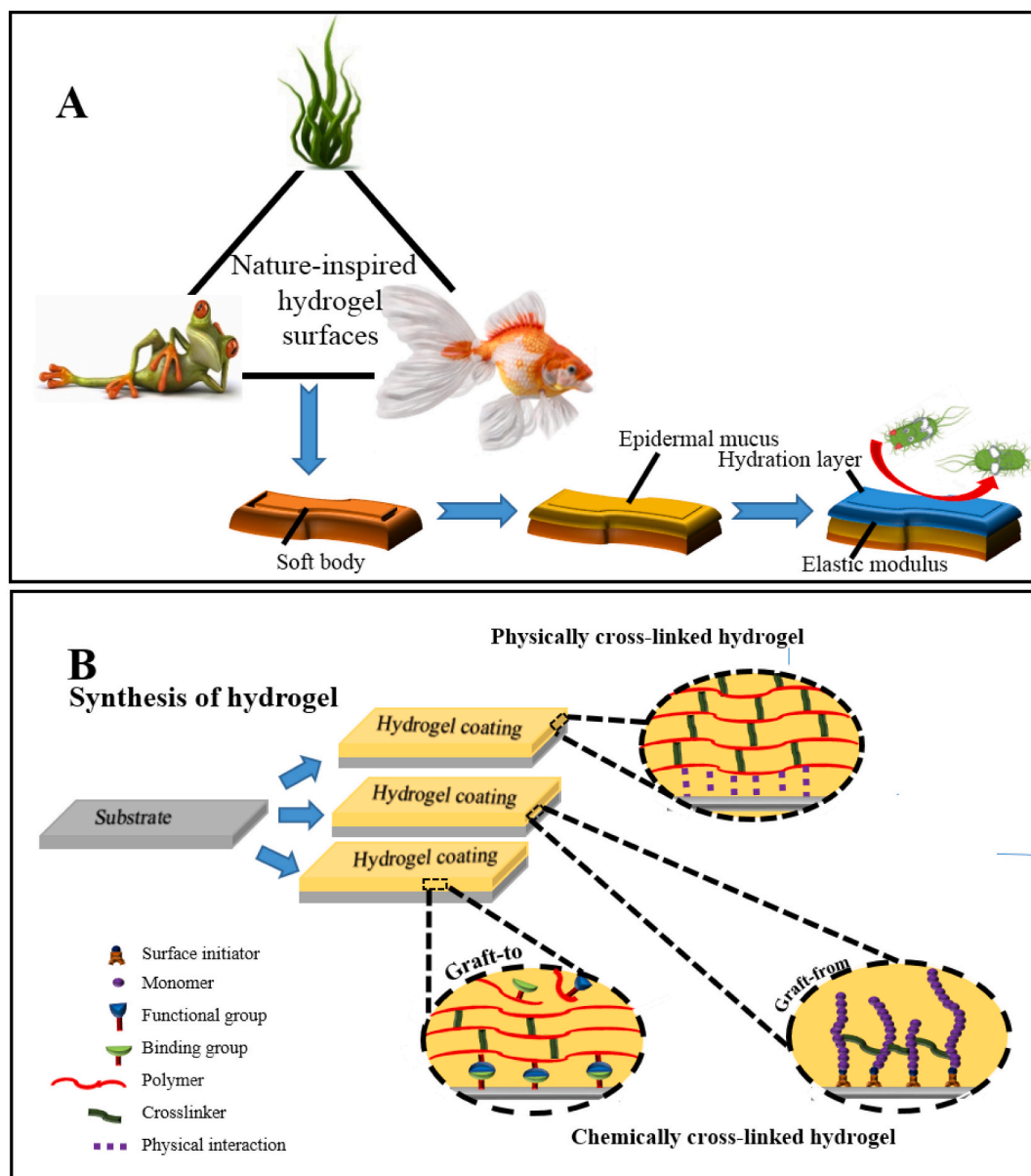


Fig. 9. (A) Schematic illustration of the hydrogel, which were inspired by living things with strong environmental adaptability. (B) Polymer-based surface functionalisation techniques employing physical and chemical processes.

superoleophilicity in the air may develop on hydrogel-like mucus and multiscale structures, which may hold water on its surface. Mucin is a natural hydrogel that is the main component of mucous [208]. Mucin is a hydrophilic protein that forms a gel in water [209]. Hydration creates a physical barrier against fouling organisms (Fig. 9A). Hydrogels resist biofouling because of their low elastic modulus. Hydrogel coatings have been extensively studied in marine AF [26,210]. The section describes the preparation of nature-inspired hydrogels and their anti-fouling activities.

(ii) Fabrication of hydrogels

Cross-linking polymers forms hydrogel through the three-dimensional porous networks. The cross-linker between polymer chains creates a network with pure elastic and viscoelastic characteristics. Hydrogel can be prepared by physical and chemical processes. Physically cross-linked hydrogels use physical contact between polymer chains to prevent disintegration, whereas chemically cross-linked

hydrogels use covalent bonds (Fig. 9B) [94,210,211]. We can briefly examine hydrogen bonding and ionic interaction to make physically cross-linked hydrogels. Chemical hydrogels are generated via covalent bonds using cross-linking agents, they are made through chemical interactions involving complimentary groups, cross-linkers, copolymerization, and high-energy radiation using multifunctional monomers.

(iii) Anti-fouling activity

Antifouling hydrogels have garnered significant interest marine antifouling research as a result of their exceptional hydrophilic properties and environmentally sustainable nature.

Natural hydrogels and synthetic hydrogels are the two types of typical hydrogels. Collagen gel, agar gel, alginate gel, and polysaccharides are biodegradable and biocompatible natural hydrogels [212,213]. Synthetic hydrogels are based on acrylate, polyurethane (PU), polyacrylamide (PAM), PEG, polyvinyl alcohol (PVA), and poly(*N*-isopropyl acrylamide) [103,214–218]. The impact of raising cross-

linking concentration on the AF efficiency of coatings was examined in the field and in laboratory testing. The coatings inhibited barnacle, diatom, and protein accumulation while improving resistance to biomass buildup. The results demonstrated that the hydrogel's AF efficiency depended on the polymers' unique chemical composition and physicochemical features like stiffness and swelling [219]. Xie et al. [220] showed that biofouling might be prevented by sequentially developing and removing a small layer of extremely swelling hydrogel on a coated surface. Wu et al. [221] discovered that the swelling ratio of cross-linked coatings steadily enhanced with immersion time. The hydrophilic groups on the coated surface might adsorb a significant quantity of water through hydrogen bonding interactions. The cross-link interaction between the polymer chains reduced water absorption as saltwater penetrated deeper. This procedure led to the creation of a thin hydrogel layer that enhanced water absorption and thickness. Hydrogels had outstanding anti-fouling properties because their high level of water inhibited proteins from constructing a link between the gel substratum to the barnacle base [109]. Recently, Yang et al. [97] reported a simple and scalable method to paint hydrogel coatings on diverse substrates using a commercial mid-coat and a specifically developed curing agent. Spray-painting and UV curing were applied to deposit the hydrogel coating on surfaces. The degradation, adhesion strengths, homogeneity, hydrophilicity, mechanical stability, and AF capabilities of the hydrogels were comprehensively tested. Many researchers have since confirmed the anti-fouling abilities of hydrogels. Moreover, hydrogels can reduce protein, algal zoospore, diatom, and barnacle adhesion [97,102,222]. Hydrogels may expand in water under osmotic pressure because they have relatively weak mechanical properties and bonding strength [221,223].

In addition, composite materials have gained popularity owing to their superior performance, low cost and ease of preparation

[75,224,225]. Hydrogels with various nanofillers, such as nano-silica, silver, and copper oxide NPs to make nanocomposites (NCs), which improve the tensile strength of the hydrogel, and NPs may be utilised as fillers in marine AF materials [223–226]. Tian et al. [227] created a variety of high-performance antifouling NC hydrogels by combining silicon substrates with Ag-NP/hydrogel (thiol/acetyl thioester chelating with Ag NPs) and to construct a novel AF surface. The compatibility of silicone polymers with hydrogels may be improved by using Ag NPs (cross-linking agents) to evenly distribute the hydrophilic polymer inside the PDMS resin matrix. The prepared coating had higher AF efficacy and became more adherent to the substrate (Fig. 10A). Two biobased hydrophilic gel coatings with exceptional mechanical and AF properties were reported by Lu et al. [228]. Hydrophilic and hydrophobic interpenetrating polymer networks (IPNs) formed in situ were used to create the coatings. Bioeugenol produced siloxane chains with bio-based epoxy monomers, which were cross-linked by isophorone diamine (IPDA) to generate hydrophobic bio-epoxy frameworks. A hydrophilic Ag NPs hydrogel network was produced via mercapto-containing hydrophilic polymer (PNIPAM-SH) and silver trifluoromethane sulfonic acid, which permeates the network of the bio-based epoxy and produces an amphiphilic hydrogel coating with IPN. The hydrophobic epoxy resin containing silicone exhibited increased tensile strength and adhesive properties, thus improving the mechanical robustness of the coating. Algae, bacteria, and other sea creatures could not survive on the hydrogel components that were cross-linked by Ag NPs and hydrophilic polymers. Conversely, the attractive force between contacting particles, induced by the large surface area of the nanoparticles, can lead to particle aggregation [229], and adversely affected mechanical properties.

However, double network (DN) hydrogels with good mechanical properties are widely used in biomedical and maritime applications. Many DN gels with an AF polymer as one of the networks have been

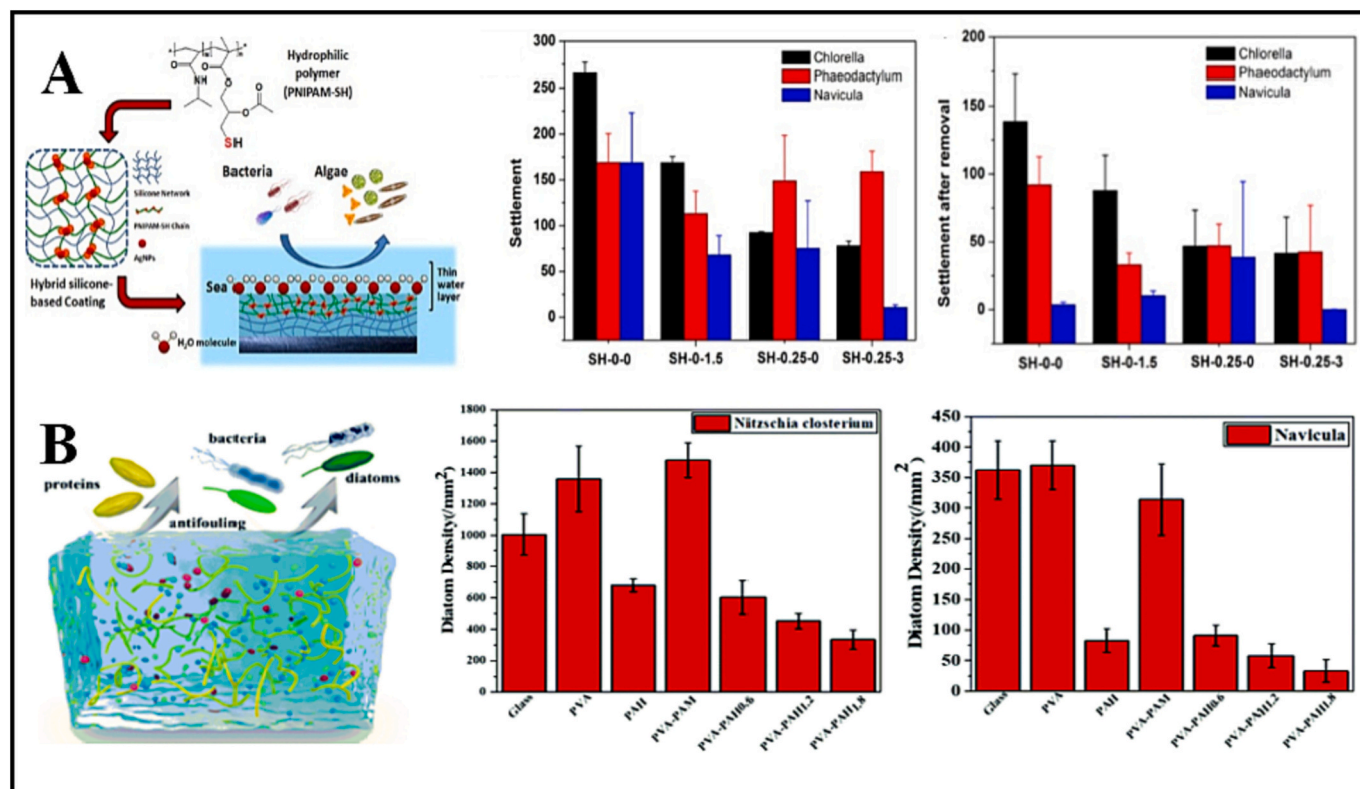


Fig. 10. (A) A novel hybrid hydrogel NC covering with silicone-based anti-fouling for long-lasting. The *Phaeodactylum tricornutum*, *navicula*, and *chloroella* settling findings on silicone-based coatings were as follows: the amount of algae adhering to the coatings in before and after water washing [227]. Figure reproduced d with permission from American Chemical Society. (B) A schematic depiction of the formulation and characteristics of the PVA–PAHX gel and the statistical representation of diatom attachment ($p < 0.01$) [94]. Figure reproduced d with permission from Royal Society of Chemistry.

shown to have good mechanical characteristics [230]. For instance, Murosaki et al. [231] conducted a long-term experiment to investigate the AF qualities of several types of hydrogels toward motile marine animals (barnacles, ascidians, and algae). The majority of hydrogels demonstrated a minimum durability of two months when immersed in the sea. The mechanically robust PVA gels and PAMPS/PAAm DN hydrogel showed AF action toward motile marine species (barnacles) for up to 330 days. The “easy-release” mechanism in which the gel’s elastic modulus and hydration layer play pivotal roles, were used to characterise the AF capability of the hydrogel against barnacles. Recently, Li et al. [94] used a capsaicin derivative, *N*-(4-hydroxy-3-methoxybenzyl) acrylamide (HMBA), to construct DN hydrogels and integrate HMBA into the polymer network. PVA is hydrophilic, and AM and HMBA may polymerize to form a brittle and hard network. They were able to generate a PVA-PAHX hydrogel with outstanding long-term AF effects, minimal swelling ratios, and high mechanical strength by modifying the quantity of HMBA in the DN hydrogels (Fig. 10B). The researcher needs to further investigate the anti-fouling properties of the DN-based hydrogel in marine field test over longer immersion times.

Hydrogels reinforced with nanoparticles are a promising class of hydrogel because they have enhanced mechanical strength and anti-fouling capabilities. Although laboratory experiments showed that the mechanical strengths of these hydrogels were increased, the test durations in marine conditions were limited. The research on nature-inspired hydrogel surfaces and its impact on biofouling development is summarised in Table 3.

4.4. Nature-inspired textures

(i) Inspired by nature

Nature-inspired texture has impact on the surface of marine structure. Many animals biological systems, such as the lotus leaf [232,233], *Salvinia molesta* [234,235], sharks [106,236–238], rice leaves [239–241], water-collecting cactus spines [242,243], geckos feet [244–246], springtails omniphobic skin [247,248], water-transporting cicada and dragonfly wings [249,250], and water-walking and leaping striders [251–253], have smart surfaces (Fig. 11). The surfaces of lotus leaves are superhydrophobic with self-cleaning qualities. “Particle-coated water droplets” slide off the lotus leaf to eliminate debris [254,255]. Springtails can breathe in water and oil because of their omniphobic skin. The non-wettability of springtail skin is explained by interconnected reentrant-shaped microstructures [256,257]. The structural composition of shark scales differs from that of scales found in other fish species. Shark scales possess a unique enamel and a diamond-like topography, although fish scales often have a flat shape resembling teeth. The distinctive functional roughness and hierarchical structure reduce adhesion and drag [236,258].

These nature-inspired textures impart surfaces with self-cleaning and AF activities, allowing organisms to adapt to their environments. Following these ideas, attempts were made to create biomimetic surfaces with AF qualities [259–261].

Table 3
Examples of recent development of nature-inspired hydrogel surfaces.

Hydrogel system	Components	Antifouling activity	Citation
Common hydrogel	Chitosan/poly (vinyl alcohol)	<i>Algae Navicula</i> and <i>Nitzschia closterium</i>	[104]
Hydrogel modified nanoparticles	Cu ₂ O-Tannic acid submicroparticles, poly (vinyl alcohol)	<i>Chlorella vulgaris</i>	[101]
Double network hydrogel	Acrylamide/alginate-Calcium/Copper-benzotriazole	<i>S. aureus</i>	[99]

(ii) Fabrication methods

The fabrication and working mechanism of nature-inspired AF texture have been extensively investigated [274]. Various approaches have been suggested to produce the nature-inspired structure with improved functionals to mimic the complex nano or microstructures. The solutions primarily involve coating and etching including bio-templating [275,276] and micromachining approaches [277].

(1) Bio-templating

The bio-templated synthesis of active compounds with well-defined architectures has several potential applications [274,278]. Sol-gel processing, chemical deposition, and nano-imprinting are some of the modification techniques.

(i) Chemical deposition: Chemical vapor deposition (CVD), electrochemical deposition technique and LbL approach, are common chemical deposition techniques.

CVD is a common surface coating technology for a wide range of applications in which a volatile coating precursor is vaporised in an inert environment by heat, light, and/or plasma discharge and then reacts on a solid substrate surface to generate a desired deposit (Fig. 12A) [279]. CVD is commonly used to create superhydrophobic surfaces since it may create low surface energy and rough structures. CVD’s high film-forming temperature and poor deposition rate are drawbacks [280]. This approach has been used to create nature-inspired surfaces mimetic of cactus and pitcher plants.

The electrochemical deposition technique is often employed to construct large-scale superhydrophobic surfaces because of its affordability, speed, and reproducibility. By adjusting charge density and electrode movement speed, uniform nature-inspired textures may be created. Electrochemical etching dissolves the anode sample inhomogeneously with current and electrolyte, producing a rough texture (Fig. 12B). Electrodeposition has been utilised to make nature-inspired surfaces, but its practical uses are restricted owing to inadequate adhesion strength and fragile micro-/nanostructures [281].

LBL deposition involves dipping an anionic or cationic substrate in an aqueous polyelectrolyte solution. The opposing charges of the substrate and solution enable polyelectrolyte adsorption on the substrate, inverting its surface charge. The substrate is then dipped in a polyelectrolyte solution with the opposite charge. These processes are repeated to make a film of anionic and cationic polyelectrolytes (Fig. 12C) [282,283]. Nanoparticles may be added for roughness. Nature-inspired lotus leaves for superhydrophobicity surfaces were made using this method [284].

(ii) Sol-gel process: The use of sol-gel in bio-templating is advantageous due to its ability to provide precise control over the nanoscale structures during the first processing stage [285–288]. In the sol-gel technique, nanometer-sized colloidal particles are dispersed in gel to create networks. Typically, the structure produced by sol-gel processing is porous. The morphology of surfaces, such as surface nanostructures, can be controlled by this strategy. The functional group, the solid surface, and the method of preparation interact to produce the final gel [289]. Transparent, thermally stable, and superhydrophobic nature-inspired surfaces have been created using this method (Fig. 12D) [290]. The sol-gel process is often inexpensive and time-consuming in commercial environments [291,292]. Furthermore, the sol-gel coating may be influenced by adhesion, cracking, and delamination restrictions [293,294]. For example, cracks and high capillary pressures occurred as a result of the drying and contraction of the wet gel, making it impossible to form enormous

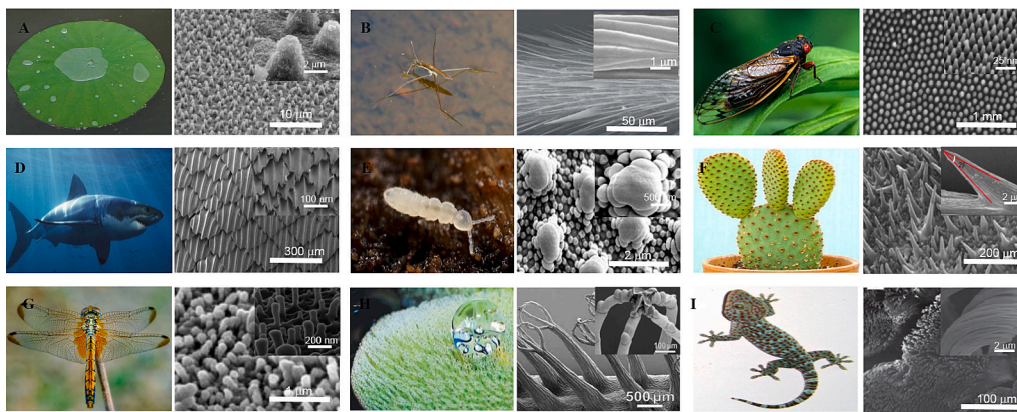


Fig. 11. Photographs and SEM images show nature-inspired textures of (A) lotus leaf (Adapted from ref. [126, 262], Figure reproduced with permission from Elsevier. (B) Water strider [263,264], Figure reproduced with permission from Elsevier. (C) Cicada [265,266], Figure reproduced with permission from Elsevier. (D) shark [267,268], Figure reproduced with permission from Elsevier. (E) Springtail [269], Figure reproduced with permission from Springer Nature. (F) Cactus [264,270], Figure reproduced with permission from Wiley. (G) Dragonfly [271,272], Figure reproduced with permission from Elsevier. (H) leaf of *Salvinia molesta* [273], Figure reproduced with permission from Wiley. And (I) gecko [264], Figure reproduced with permission from American chemical Society.

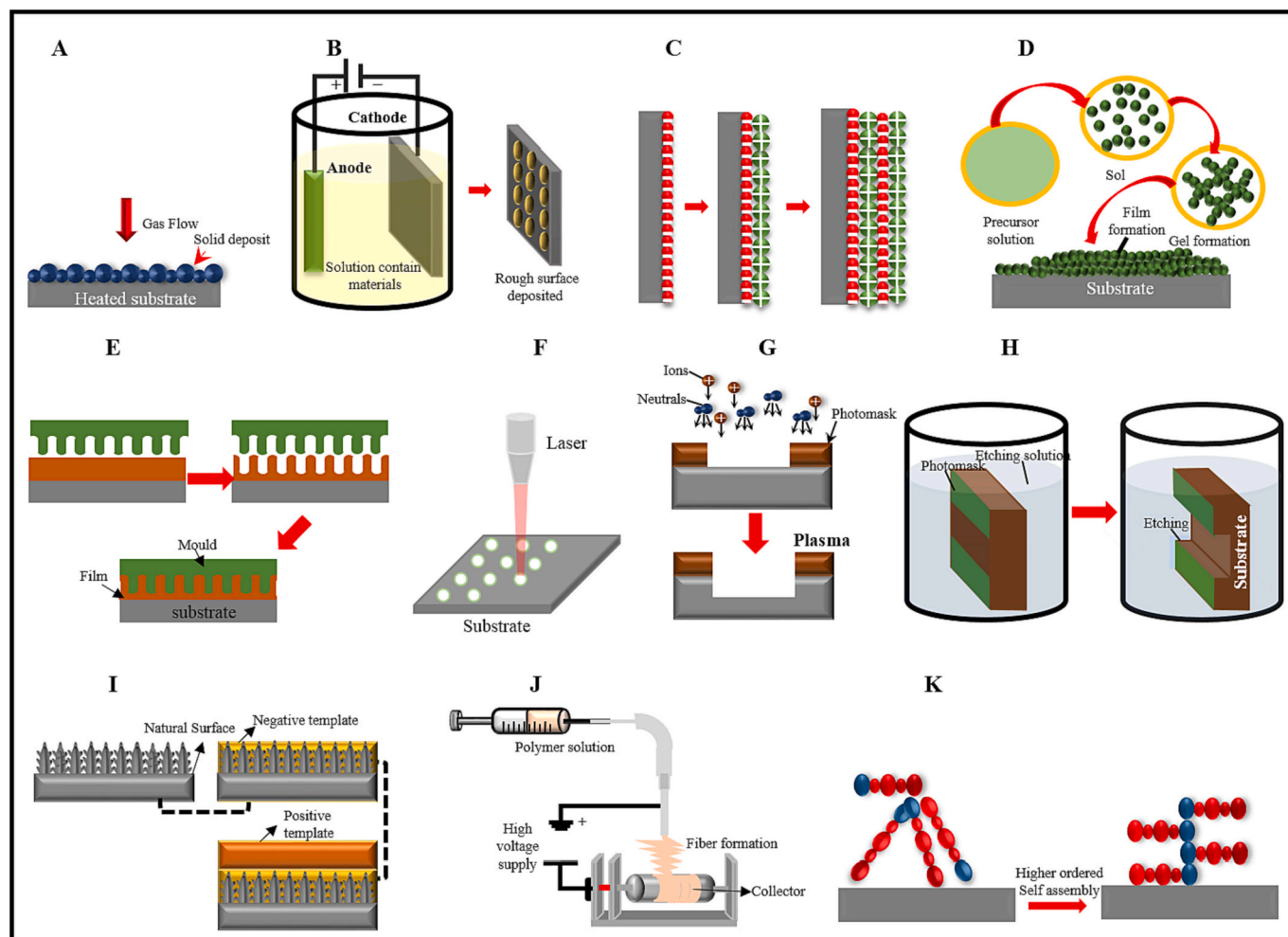


Fig. 12. Schematic illustrations of major nature-inspired texture fabrication methods, including (A) CVD, (B) electrodeposition, (C) LBL deposition, (D) solgel, (E) nanolithography, (F) laser etching, (G) plasma etching, (H) chemical etching, (H) electrospinning, (I) templating, (J) electrospinning, and (K) self-assembly.

blocks. In addition, insufficient drying or hardening may produce organic or carbon residues that decrease coating performance.

(iii) Nano-imprinting lithography: The nano-imprinting technique, also known as nano-imprinting lithography (NIL), is a pattern-

creation process that uses mechanical deformation of imprint resist. In particular, polymerizable or polymers monomers imprint resists hardening by UV radiation or temperature stimuli during imprinting, ensuring adequate adhesion between the

templates and resist. (Fig. 12E) [295]. Several derived NIL methods exist, such as resist-free direct thermal NIL, photo NIL (P-NIL) and thermoplastic NIL (T-NIL). NIL has been utilised to create nanopatterned polymer anti-fouling surfaces that resemble the nanopillars in dragonflies and cicada wings [296]. Using this technique, antireflective, self-cleaning, and superhydrophobic surfaces that are inspired by the eyes of moths and the wings of a cicada have been created. The use of pattern replication technology in NIL offers a substantial reduction in costs compared to the use of costly optical imaging and photomasks equipment. NIL is one of 10 new technologies that are expected to revolutionise the world as a viable next-generation lithography technology [297].

(2) Micromachining

The micromachining technique, which utilises the concept of bio-structural materials, may be classified into top-down and bottom-up approaches. Top-down techniques include lithography, etching and soft-template method [298,299], whereas bottom-up technique primarily comprises electrospinning and self-assembly [300,301]. Due to the intricacy of nature-inspired surfaces, several methods ranging from basic to compound have been constructed over decades to obtain the bio-structure in artificial preparation. Several micromachining approaches have shown promise in overcoming the limitations of templating methods.

- (i) **Lithography:** The terminology lithography derives from the Greek words: lithos (stone) and graphia (writing) [302]. Lithography techniques enable the rapid fabrication of irregular or regular patterns at the micro- or nanoscale level. Typical lithography method involves photoresist depositing/coating, exposure, development, transfer, etching and stripping [303]. For the purpose of creating morpho structures, a technology referred to as laser interference lithography (LIL) was developed [304]. The LIL required three steps: sample preparation, exposure, and development. Comparing to the conventional LIL method, a reflective optical coating was applied to the fused silica substrate prior to the deposition of positive photoresist (AZ 1505) on the sample for Morpho structure fabrication through sputtering. The surface was exposed via a two-beam LIL that originated from a single laser source, which results in high coherence. Subsequently, the application of the developer produced the tree-shaped structure. Different metals, such as Zn, Fe, Cu, Ni, Ti, and SS, may also be given groove structures, which display spectacular structural colour and anisotropic wettability [305]. This is in addition to the photoresist. It is intriguing to notice that the groove duration was at the micrometer scale; the structural metals' surface wetting relied on the groove height. LIL is cheaper than optical lithography since it requires no mask or clean room (Fig. 12F) [304].
- (ii) **Etching:** Plasma etching is a dry cleaning process [306,307]. In plasma etching, high-energy reactive species from introduced gases, such as NH_3 , CF_4 , O_2 , or Ar, approach and react with the sample surface (Fig. 12G) and the surface becomes rough [308]. Plasma etching is simple and reproducible; and can be applied to metals, alloys, and polymers surfaces [309]. This approach has examined lotus, gecko, cicada wing, dragonfly, and moth microstructures [310]. Chemical etching is a wet procedure that can be used to fabricate micro-/nanostructures using acidic or basic solutions [311]. It has been used to create nature-inspired surfaces (Fig. 12H) [312,313]. Chemical etching works on alloys and metals that may interact with the etchant. Inert polymers like polystyrene and silicone are not etched to create nature-inspired surfaces.
- (iii) **Soft-lithography Method:** Templating, often termed as soft lithography, mimics natural texture as closely as feasible. This is a

common way to make natural-textured replicas [237,314]. The templating process involves two parts (Fig. 12I): making a negative mould from positive reproductions and natural surfaces [315]. This technology has the ability to replicate natural surfaces with ease. In certain cases, the dimensions of natural surfaces may prove insufficient for the creation of extensive negative moulds. Laser ablation, electroplating, chemical etching, and computer numerical control (CNC) machining have been used to make negative moulds [316–318]. Templating method is promising approach for the mass production of soft materials [319]. Lotus leaves [320], cicada wings [321], and butterfly wings [237] are examples of nature-inspired surfaces created utilising this template process in 1D, 2D, and 3D.

- (iv) **Electrospinning:** Electrospinning is a technique for rapid production micro or nanoscale nature-inspired threads and fibres by viscoelastic liquids in an outside electric field (Fig. 12J) [322,323]. For the purposes of a laboratory experiment, it is necessary to have a high-power supply, a syringe pump, a conducting collector, and a polymer with a high molecular weight. The electrospinning process is initiated by applying an electrical potential between a syringe nozzle and a conducting collector, which leads to the formation of a high electric field. In general, the polymer solution is expelled to generate a charged liquid jet from the apex of a deformed droplet, as a result of the electrostatic repulsion between charges caused by the elevated electrical potential, which overcomes the surface tension. Afterwards the occurrence of whipping and bending instabilities along the syringe nozzle to conducting collector trajectory, the liquid jet is afforded sufficient time to allow for complete solvent evaporation and subsequent deposition onto the collector. This process ultimately culminates in the formation of a solid fibre membrane [322–325].
- (v) **Self-assembly:** Self-assembly effectively produces ordered nanostructures from fundamental structural components, such as nanoparticles, molecules, and atoms [326,327]. The molecular structural arrangement may affect the macroscopic characteristics of materials in self-assembly procedures (Fig. 12K). The design may be generated on the whole substrate at the same time. This approach has created nature-inspired surfaces mimicking that of morpho butterfly wings with structural hues, self-cleaning, or water-repellent [327,328]. This approach has also been used to create nature-inspired surfaces, that mimic green bottle fly eyes, lotus leaf and rice leaf [311,329].

(iii) Anti-fouling activity

Several creatures, such as sharks, sea crabs, lotus leaves, and rice leaves, possess remarkable self-cleaning and antifouling properties on their surfaces. These properties efficiently prevent or hinder the attachment of fouling organisms.

Non-bactericidal AF coatings featuring nature-inspired topographies of marine creatures, such as sharks might prevent them from invading organisms. The shark skin is formed of structured dermal denticles surrounded by riblets, and its topography decreases drag flow and prevents microorganism settlement [330]. Thus, surface topography effects on adherence of cells and microorganisms, and minimises of the biofouling development. The surface texture of sharkskin reduces the settling of marine creatures (Fig. 13A). The efficacy of biofouling prevention has been related to the surface mucous layer, cutaneous denticle bending, and microstructured riblets.

In the marine environment, the variations in dimension among target organisms cause eradication marine fouling to be a stressful process. As such, it is still a problem to improve long-term and broad-spectrum AF effect by mimicking the topography of shark skin [331–333]. Sharklet, a commercial AF surface which is inspired by shark skin [334,335]. Munther et al. [336] created Sharklet-patterned surfaces with an

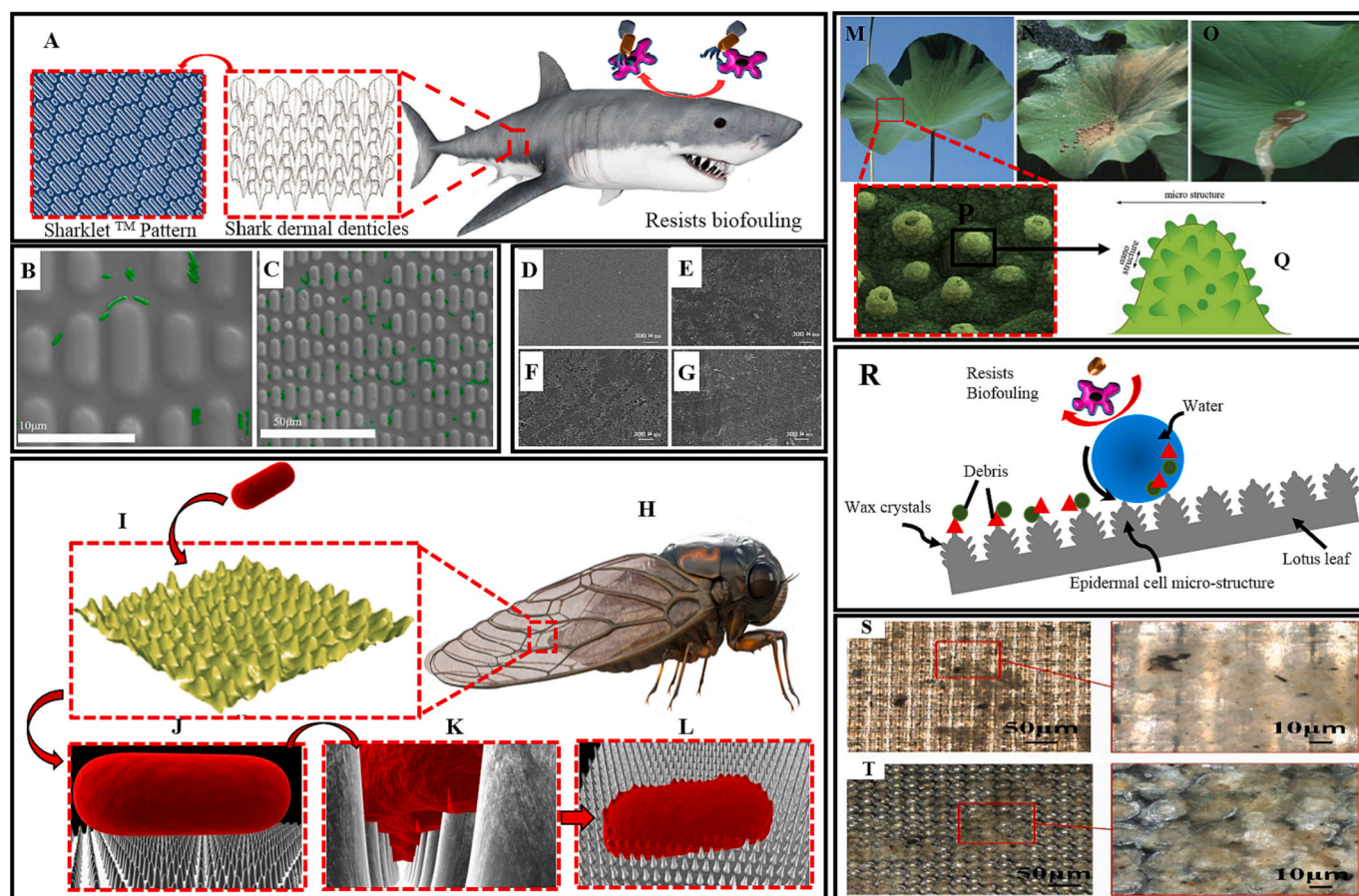


Fig. 13. (A) Sharkskin-mimetic Sharklet™ micropatterned surface, which reduces biofouling in the marine environment, is shown in best perspective. (B–C) SEM picture depict the settlement of on surfaces are comparable characteristics, lower magnification displaying a variety of structures and the subsequent bacteria colonization. Scale bars in and out are (B) 10 and (C) 50 μm , respectively [336]. Figure reproduced with permission from Elsevier. (D–G) Construction of a nature-inspired shark skin surfaces with anti-biofouling and drag-reduction capabilities [337]. Figure reproduced with permission from The Company of Biologists limited. (H, I) Cicada-inspired antimicrobial activity, (J–L) The nanopillar-covered surface that stretches and tears bacterial membranes belongs to the mechanically processes of nanostructured surfaces. (J) The surface of a nanopillar on the wing of a cicada kill the microorganism. (K) The membrane is stretched and adheres to the surface protrusions. (L) Cell lysis finally results from continued membrane adsorption, which significantly stretches the cell membrane hanging between each nanopillar [342]. Figure reproduced with permission from Elsevier. (M) Sacred lotus-inspired self-cleaning. (N, O) Rinsing water easily removes dirt [343]. Figure reproduced with permission from Springer Nature. (P, Q) Microstructure/nanostructure of lotus leaf surface [344]. Figure reproduced with permission from Elsevier. (R) Schematic illustration of the tilted posture of a lotus leaf causes water droplets to slide off [345]. Figure reproduced d with permission from Elsevier. (S, T) Photographs of Super hydrophobic surface after about five weeks of immersion. (S) Super hydrophobic surface with micro-groove pattern; (T) Super hydrophobic surface with micro-pit pattern [260]. Figure reproduced with permission from Elsevier.

integrated height gradient to further reduce organism adhesion. This design did not only reduce the number of adhesion sites, but the difference in height between surrounding parts might also prevent organisms' adhesion within the spaces (Fig. 13B, C). In another study, Fig. 13D–G shows a comparison of micro-textured shark-skin replica and a smooth PDMS replica made from materials that were immersed in China's Pearl River for 70 days before being removed and cleaned. The shark-skin replica surface exhibits improved anti-fouling properties against algae species [337]. Shark-skin surface topology has been shown to have superior AF properties when compared to micro-grooved or smoothed textures, even when created on the same material [338,339]. Micanti [340] designed biomimetically-textured coatings made with nylon fibre (Thorn-D1) for use on ships, aimed at reducing fuel consumption and improving anti-fouling efficiency [341]. Research shows that sharkskin biomimicry on exposed hull surfaces can save energy, increase velocity (reduce drag) and boost anisotropic flow in turbulent water flow. Silicone textured surfaces resembling sharkskin topology have reduced drag by over 15 % on coated vessels and submarines and have decreased algae cell adhesion by up to 67 %. Recently, Li et al. [106] developed an iron-based amorphous coating with Cu_2O infused

into the nanofibers, which produced a superhydrophobic surface with bactericidal-resistant effect due to supply of Cu^+ ion. The coating was found to have outstanding antifouling properties, with quantifiable resistance to *P. aeruginosa*, bovine serum albumin, and *N. closterium*, respectively.

Some land-flying insects, such as dragonflies and cicadas, have superhydrophobic wing structures that resist microbial adherence and nanostructured surfaces also reduce organic pollutants under immersion due to hydrodynamic pressure. The nanostructured capabilities may tear through bacterial cell membranes when they adhesion to the surface, resulting in death of cell in marine fields as well as biomedical [346,347]. The mechanically anti-fouling process has been theoretically modelled [348] which is based on the observation that nanopillar surfaces in nature exhibit broad-acting anti-fouling properties. Membrane rupture develops in the suspended areas within nanopillars when the bacterial cell adheres to the nanopillar surface (Fig. 13H–L) [342,349]. The mechanism of mechanically anti-fouling involves the bacterial membrane being stretched exceeding the elastic threshold. Since 2012, several studies on natural bactericidal surfaces have been conducted. The cicada wings with a uniformly dispersed nano-needle array on their surface may kill Gram-

negative bacteria in 15–20 min via immediate physical contact, as previously documented by Mainwaring et al. [350]. However, the animal surface has a lower water CA than the plant surface, and there is no discernible variation in the anti-fouling effect [351], which showed that the linkage between hydrophobicity and bactericidal efficiency, is not particularly close [352]. Kelleher et al. [353] discovered hydrophobic nanostructures with low surface energy on the wings of three distinct cicada species. Ormstamp surfaces killed most Gram-positive *S. aureus* cells with 40 pillars/m². Cicada wings exhibit species-dependent bactericidal effects [353]. A biophysical model showed that cell membrane stretching in the portions dangling between the pillars destroys the cell in the encounter [348]. It has been hypothesised that nanopillars may tear bacterial membranes like a “bed of nails” [249,354]. A biophysical model [355] predicts that increased pillar density would increase stretching points, and the close proximity of the pillars will be reduce the bacterial membrane’s interaction with the vertical pillar walls. This hypothesis suggests that nanocone surfaces and inhomogeneous height pillar arrays may stretch bacterial membranes [356]. The surface material’s chemical properties may affect the pillar walls’ contact with the bacterial membrane. The mechanical capabilities of various bacterial cell membranes, which produce additional shear pressures through cell motility or flow, and substrate. Heckmann et al. [357] found the aforesaid structure-property connection. *S. aureus* and *E. coli* are prevented by PDMS nanopillar arrays with 480 and 1100 nm pitches, respectively. The surface structure of specific nanopillar arrays (e.g., 480 nm) may be antibacterial against *E. coli* and anti-fouling against *S. aureus*. Atomic force microscopy (AFM) was utilised in another study to demonstrate that these nanopillars may pierce bacterial membranes and destroy cells in under 3 min [358]. These nano topographies destroy Gram-negative bacteria such as *P. fluorescens*, *E. coli*, and *P. aeruginosa*, but not Gram-positive bacteria due to stronger cell walls. Gram-negative and Gram-positive bacteria were drastically decreased up to 99.3 % at 450,000 cells/min/cm² through nano-features on vertically aligned black silicon and carbon nanotubes, according to research by Ivanova et al. [22] and Linklater et al. [359]. Nanostructures comprising nuggets, rings, and pillars exhibited >99 % decrease of *S. aureus* [360]. Furthermore, Gangadoo et al. evaluated the AF efficacy of nanostructured wings of dragonfly, while the most chaotic surface had the highest fouling resistance than smooth surface. This was suggested that air pockets cause poor adsorption properties compare to lotus leaf surface [112,361].

In addition, several plant leaves have outstanding water-repelling properties [362,363]. The lotus leaf is the most common example that has inspired material researchers to create super-hydrophobic multifunctional surfaces for a variety of commercial applications. Physical and chemical qualities like hydrophobicity are related to surface roughness. Surface wettability (surface-free energy) is linked to tiny protrusions that contribute to partial wetting (how effectively a liquid can coat a surface and remain in touch with it). Micro-nano-features inhibit the attachment of bacteria by entrapping air (bubbles) between the surface topography and producing an anti-biofouling effect (Fig. 13M–R) [364–366]. Microscale air surface protrusions are low adhesion and super-hydrophobic, which allow water droplets to slide off pollutants. The term ‘lotus effect’ refers to droplets of water rolling off the water-repellent surfaces of lotus leaves (*Nelumbo nucifera*) [367]. In liquid, microorganisms cannot permeate the air-liquid barrier due to strong interfacial tension. It makes it challenging for biofouling to locate anchoring points on surfaces with a wide and robust air-liquid interface. Super-hydrophobic coatings have the potential to hinder the adsorption and development of microorganisms on coatings throughout the peri-operative period. This is achieved by preserving an air layer that reduces the available surface area for cell attachment [368,369]. Using the anti-fouling properties of lotus leaf as inspiration, the coating substrate with super-hydrophobic materials is a solution to the issue of biofouling. At this stage, air pockets trapped between the solid and liquid phases prevent direct contact between solid surfaces and biological organisms. Therefore, biofouling is unable to adhere to the substrate. Sun et al.

[260] used a picosecond laser to produce superhydrophobic surface with hierarchical micro/nano texture on a SS substrate. Anti-biofouling surfaces have decreased 50 % microorganism adhesion compared to untreated surface (Fig. 13S–T). Zhang et al. [370] and Hwang et al. [371] indicated that using super-hydrophobic surfaces with anti-biofouling properties might substantially impact the development of protein, bacteria, and marine creatures. Super-hydrophobic anti-fouling compounds may be used for ship hulls, submarines, and undersea structures. Numerous super-hydrophobic surfaces were manufactured using other substances than PDMS. Furthermore, these hierarchical textured surfaces demonstrated remarkable anti-fouling characteristics in field and laboratory measurements. Efimenko et al. [372] created two coatings with identical chemical compositions. The surface of one was smooth, whereas the surface of the other was wrinkled in a hierarchical manner. Marine field experiments concluded that the hierarchically wrinkled surface could only be maintained clean for 18 months compared to 4–15 weeks for the flat surface. A relatively recent finding is the potential of lotus leaf surfaces to kill cells directly by destruction the membrane. As recently published by Jiang et al. [373] these results have inspired scientists to create surfaces that integrate nano- and micro-scale features with per-fluorination for a more significant anti-biofouling synergistic impact. These surfaces demonstrated >99 % attachment resistance against *E. coli*. These outcomes showed that the lotus mimic surface is better than other surfaces, because they exhibit superhydrophobicity and mechanical contact-killing behavior, respectively.

Most data collected and published in scientific articles do not comply with a uniform set of surface characterisation criteria that permits direct comparisons over several studies. These observations have prompted attempts to develop materials with an optimal surface roughness design that mechanically kills and repels biofouling investigated in marine environments. However, scientists recently identified an unexpected phenomenon in which lotus leaves exhibited an innate bactericidal effect toward adherent microorganisms. They also discovered that the antibacterial action might be attributable only to physical nanostructures rather than surface chemicals. However, surface topology alone may not be adequate for long-term biofouling management. Investigations of marine species’ defensive systems lead researchers to conclude that the combination of topography and chemical composition will be crucial for acquiring the appropriate surface attributes. The research on nature -inspired textured surfaces and its impact on biofouling development is summarised in Table 4.

4.5. Extracted biological macromolecules (EBM)

Recently, researchers have been searching for biological macromolecule AF compounds to replace chemical anti-fouling agents. Biological macromolecule made from natural bioactive compounds extracted from terrestrial and marine organisms have AF action [375–378]. The chemical compositions and structures of natural products may be used to generate innovative marine and biomedical AF coatings, respectively (Fig. 14A) [379–381].

Table 4
Examples of recent development of nature-inspired textured surfaces.

Main components	Inspired by	Antifouling activity	Citation
PDMS	Sharkskin	<i>Chlorella vulgaris</i>	[339]
Silica hollow colloidal crystal/ polyperfluoroether acrylate- polyurethane acrylate	Sharkskin	<i>E. coli</i>	[374]
Polyurethane	Lotus	Protein and protozoan	[111]
PDMS	Cicada and dragonfly	Protein, carbohydrate and DNA absorption	[112]
Poly (ethylene terephthalate) and TiO ₂ NPs	Sharkskin	<i>E. coli</i> and <i>S. aureus</i>	[236]

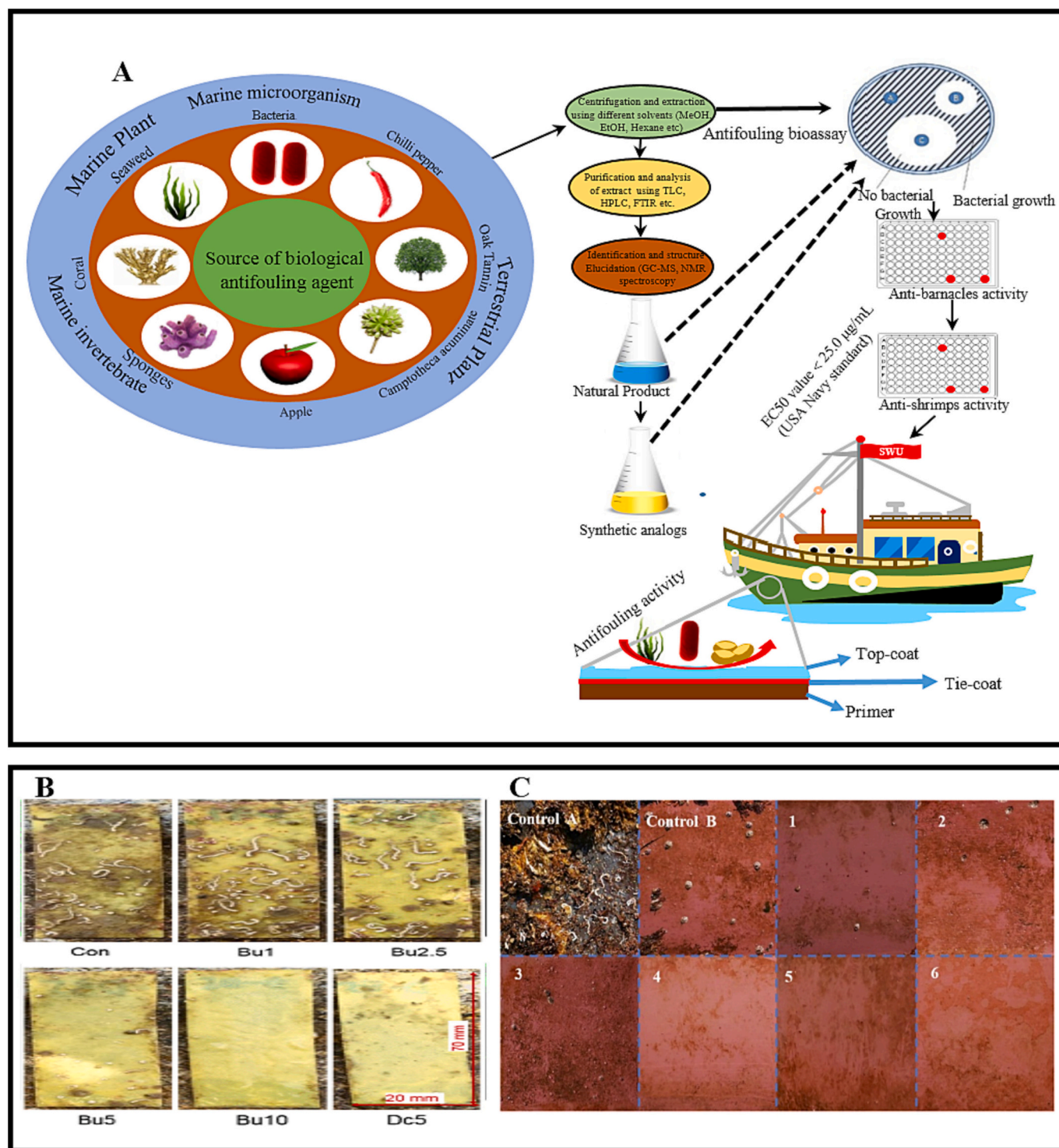


Fig. 14. (A) Schematic Illustration showing the evolution of natural product and their synthetic analogs. (B) The AF performance of surfaces coated with butenolide and polyurethane with a poly (-caprolactone) basis (1, 2.5, 5 and 10 wt%) at various concentrations. Direct imaging was used to monitor the anti-biofilm test after it had been running for 36 days in a subtidal environment [382]. Figure reproduced with permission from Taylor & Francis. (C) The result of anti-fouling coating are after three months in the water. The blank group was not coated with anti-fouling paint, whereas the control group was treated with anti-fouling paint without capsaicin derivatives, 1–6 identification to compound A-F 10 [383]. Figure reproduced with permission from Elsevier.

(i) Anti-fouling activities

(1) EBM from marine organisms

Many BM can be produced from marine invertebrates and plants, such as sponges, seaweed and bacteria. These substances are safe for the

maritime environment and have specific anti-fouling properties [384–386]. These compounds have AF properties and are nontoxic to the marine ecosystem. This section will thoroughly discuss typical anti-fouling agents derived from marine organisms.

Special bioactive metabolites generated by sponge-associated biota or sponges are essential for their existence in aquatic environments. As a

consequence, they may have antibacterial and AF capabilities [387,388]. Sponges naturally manufacture AF substances to prevent marine larvae and bacterial strains from sticking to their bodies, which stop to filtering activities [389,390]. A bile acid derivative known as peracetylated cholic acid was extracted from the Patagonian sponge *Siphonochalina fortis*. The addition of sponge chemicals are added into soluble-matrix paints and subsequent testing in the sea. *M. edulis platensis* settlement experiments demonstrated AF action and the highest concentration to eliminate 50 % of the mussel population [116].

Marine plants (seaweed) show anti-bacteria and antifouling performance. Elangovan et al. [391] evaluate the anti-fouling properties of crude organic extracts derived from five different species of seaweed obtained from the Rameswaram coastal region at Pamban station, Tamil Nadu. A collection of 15 distinct compounds were examined, and each molecule had different retention periods ranging from 21.70 to 34.49 g/mL. The methanol extract of *P. hornemannii* exhibited the highest level of inhibition in byssal thread formation in the green mussel species compared to chloroform extract (*P. hornemannii*). The current investigation has determined that the methanol extract derived from *P. hornemannii* exhibits promising secondary metabolites, thus making it a viable candidate for large-scale production of antifoulants.

Actinobacteria is the biggest phylum of bacteria and has considerable potential for creating bioactive natural compounds, including butenolide [392]. Butenolides are extracted from two actinobacterial strains, such as *Streptomyces* sp. (GWS-BWH5) and *Streptomyces albidoflavus* (UST040711-291). The compound of butenolide hindered *B. amphitrite* larval settling (<10.00 g/mL); however, their toxicity was low [393,394]. AF bioassays used a dozen butenolide analogs because of their high AF activity. Boc-butenolide modified with a Boc-protecting group (termination of the alkyl group) has comparable AF capabilities to butenolide in larval-settling bioassays. The melting point of boc-butenolide is 132 °C as compared to butenolide's 23 °C. In a marine field experiment, boc-butenolide coatings had limited biofouler coverage after two months [395]. Ding et al. [382] recently discovered the anti-biofilm activity of butenolide against marine microorganisms. After one month, panels coating with 10 % butenolide paint were devoid of biofilm development, while only a minor quantity of biofilm formed on 2.5 and 5 % panels (Fig. 14B).

Marine biological antifouling agents cannot be manufactured commercially due to difficult extraction procedures, high prices, and an inability to get significant quantities from the ocean. The problem with these chemical components practical usage is that they do not have strong broad-spectrum anti-fouling properties for long time in marine field. The research on EBM from marine organism and its impact on biofouling development is summarised in Table 5.

(2) EBM from terrestrial plants

Table 5

Examples of recent isolated EBM from terrestrial plant.

Extracted chemicals	Source	Target	Citation
Cardenolides odoroside A, digitoxigenin oleandrin and odoroside H	<i>Nerium oleander</i>	<i>Barnacle balanus albicostatus cyprids</i>	[415]
Capsaicin	Capsicum	<i>P. tricornutum</i> , <i>Skeletonema costatum</i> and <i>Chaetoceros curvisetus</i>	[120]
Indole and furoquinoline alkaloids	Guatambu tress	<i>Mytilus edulis platensis</i>	[416]
TJ-2 (extracted from fruit) and CT-6 (extracted from leaves)	<i>Zanthoxylum bungeanum</i>	<i>Navicula</i> sp. <i>Amphora</i> sp. and <i>porphyridium</i> sp.	[417]
Terpinen-4-ol	<i>Melaleuca alternifolia</i>	<i>S. aureus</i>	[418]

Numerous investigations have been done on organic antifoulants produced from terrestrial plants, such as chili peppers, the camptotheca acuminate tree, and other plants.

AF action has been found for various plant tannins. Tannin-rich plant isolates have strong antifouling properties. The differences in AF effectiveness are based on action time, solvent/matrix type, temperature, and pH [396]. Tannins are polydentate ligands that attach to proteins mainly through hydrogen bonding and hydrophobic interactions [397]. It was revealed that bacteria did not damage leather objects in shipwrecks, because they are remarkably protected on the bottom of the hull. Tannic acid is a polyphenol-protein complex [398–400]. Tannic acid was isolated from the black wattle tree and combined with a soluble material to create an anti-fouling coating by Peres et al. [375]. The AF experiment demonstrated that the coating has a considerable impact for 7 months, which confirm the excellent anti-fouling efficacy of tannic acid. Chalcones are flavonoids in many fruits (e.g., tomato, apple, citrus) and vegetables (e.g., potatoes, bean sprouts, shallots). They contain antibacterial capabilities and may cause a variety of biological actions. Sathicq et al. [401] synthesised six chalcone derivatives and tested them against marine algae and calcareous tubeworms for anti-fouling activity. They produced furyl-chalcone anti-fouling coatings using furan rings instead of B rings (phenyl) and tested them in the marine environment. Experimental results show that furyl chalcones reduce marine fouling. Furyl-chalcone is biodegradable and has minimal influence on the marine environment, and its production is straightforward and cost-effective. However, the mechanism of action of chalcone derivatives as antifoulants is not well known. Therefore it is crucial to investigate the toxicity of furanyl chalcone to non-target marine creatures prior to using it in the marine environment. Alkaloids are bioactive compounds that are produced by many terrestrial plants. The anti-fouling properties of 18 different alkaloids were investigated by Feng et al. [402]. Four of them with camptothecin (CPT) having the highest fouling resistance. The coating using CPT as an antifoulant outperformed commercial metal-based coatings during a 12-month field experiment. CPT was also less harmful than various commercial antifoulants, such as *Isochrysis galbana*, *Phaeodactylum tricornutum*, planktonic algae, and brine shrimp *Artemia salina*.

An alkaloid in the fruits of the capsicum genus (Solanaceae plant family) called capsaicin (8-methyl-N-vanillyl-6-nonenamide) is responsible for the pungency of almost all peppers, such as chili peppers. Capsaicin is good in preventing biofouling with ecologically friendly. Buchholz first partly isolated capsaicin in 1816, and Thresh subsequently named it after isolating the pure crystalline form [384,403–406]. The global output of chili peppers is adequate for its uses, reducing its cost. The most used technique uses ethanol, chloroform, and hexane for solid-liquid isolation [407]. Capsaicin and its derivatives have indirect and direct AF effects, which are relying on experimental methods, bacterial species, strains, chemicals, and doses [408–411]. Peng et al. [412] studied anti-fouling effect of capsaicin. The marine field test demonstrated that paints containing capsaicin concentrations ranging from 1 % to 5 % exhibited anti-fouling action for a duration of 18 months. The anti-fouling function of the paints was completely diminished after a period of 21 months as a result of leaching. Capsaicin has been added to organic coatings to improve antifouling capabilities. A capsaicin-modified acrylate polymer composite showed antifouling activity [413]. Wang et al. [120] created six capsaicin derivatives and evaluated them for toxicity and fouling resistance to various types of algae. The inhibition of fouling was 95 %, while the toxicities were mild to moderate. Capsaicin derivatives are thought to act as algacide by destroying the membrane structure of algal cell. In marine field tests, the derivatives of capsaicin showed good AF performance for at least 3 months. Recently, Xu et al. [383] used the Friedel-Crafts alkylation procedure to create six capsaicin derivatives, and their anti-fouling properties were investigated against bacteria, algal as well as practical panel experimenting in a marine field. All capsaicin derivatives suppressed bacteria (*Pseudomonas* and

Flavobacterium) and algal development (*Nitzschia closterium*, *Chlorella sorokiniana*, *Chaetoceros curvisetus*, *Skeletonema costatum*, and *Phaeodactylum tricornutum*). The compound had the most remarkable algal inhibition rate (88.34 %) and antibacterial rate (93.05 % at 3 mg/L). Researchers discovered that capsaicin compounds having lower carbon in the amide side chain inhibited algae. After 7th days, marine field testing revealed that some algae had attached to the coatings (Fig. 14C). Other research [117,414] corroborates that the marine environmental risk associated with capsaicin is negligible.

There are many different sources of biological antifoulants, and the results are quite good. However, the implementation of the approach for species removal is now faced with significant challenges in terms of scalability. Although the synthesis of some physiologically active compounds is possible and this approach is far less feasible compared to metal groups. Additionally, natural product anti-fouling agents may exhibit ecotoxicity, which may be dangerous to creatures other than the targets organism. The research on EBM from terrestrial plant and its impact on biofouling development is summarised in Table 5.

(ii) Synthetic analogs and challenges to identify their suitability

Natural antifoulants are good for the environment, but several number of factors limit their development. The collecting of natural product from plants and microbes are sustainable practice. Furthermore, natural antifoulants could not show functionality as well against fouling as hoped. As a result, the synthesis process of these molecules is potentially viable for their commercialization strategy. However, natural products have an intricate chemical composition, which are challenging to produce in sufficient quantities with reasonable price [381]. Therefore, simpler analogs are preferable for use in the future study, and natural materials serve as sources of inspiration instead [381,419]. A long-term ecological risk assessment must approve any chemical prior to commercialization because of the tight regulations and administrations [66,420]. However, few toxicological analyses of antifoulants exist for both target and non-target species [381]. Because laboratory studies are often utilised in research and are inadequate and confined to a small number of fouling species, it is impossible to identify the broad-spectrum effectiveness of antifoulants [381]. Therefore, the investigations of anti-fouling compounds' mechanism of action should be carefully considered [381,420,421].

Butenolide, an effective, nontoxic anti-larval settling agent, was synthesised to tackle three of the most important fouling species and also reduced biofoulers by 70 % on marine paint. The butenolide has a simple molecular structure, low levels of toxicity, and demonstrates a strong antifungal (AF) activity. These characteristics make it a promising candidate for incorporation into AF formulations [395,422]. In this regard, the acute toxicity of butenolide on microalgae, crustaceans, and fish, as well as its projected no-effect concentration (PNEC). The predicted no-effect concentration (PNEC) of butenolide was determined to be 0.016 mg/L, surpassing the values observed for other recently developed biocides. Butenolide-activated JNK, proteasome/lysosomes, caspases, and Bcl-2 family members in *HeLa* cells to trigger apoptosis in zebrafish cells and embryos [423]. Marine medaka (*Oryzias melastigma*) were utilised in complete toxicity research of butenolide. Butenolide was also tested on *A. amphitrite* cyprids larvae and *B. neritina* swimming larvae. Both larval species' positive phototactic behavior was suppressed ($EC_{50} = 0.82$ and 3 mg/mL, respectively). Butenolide increased in the adhesion of *B. neritina* larvae to the experimental wells and also affected on *A. amphitrite* cyprid attachment due to temperature. The adherence of *B. neritina* and *A. amphitrite* was impeded by the application of butenolide at concentrations of 4 and 10 μ g/mL; however, the affects were reversible after a specific exposure duration [424]. Butenolide protected field-coated surfaces for three months. Biofouling on submerged surfaces was unavoidable after three months. Thus, multiple butenolide-containing paint formulations were tested to see how to paint components affected field performance.

Terrestrial plants are simpler to harvest than sea species. Therefore, Pérez et al. [425] synthesised (7-hydroxy-4-methylcoumarin)

compounds having AF action. Under controlled laboratory circumstances, compound 7-hydroxy-4-methylcoumarin's anti-fouling ability was assessed with the bivalve *Mytilus edulis platensis* (EC_{50} value of mussel attachment: 11 μ g/cm²), which successfully prevented the settlement and biogenesis of mussels. During a 90-day duration of marine panel testing in real-world conditions, it was shown that coatings containing 7-hydroxy-4-methylcoumarin exhibited significantly reduced fouling coverage. Furthermore, It is important to remember that capsaicin are human treasures [404]. The outstanding anti-fouling activity of capsaicin analogs against biofouling organisms in maritime field experiments has also gained a lot of interest recently [426]. Because capsaicin may kill bacteria on contact with a surface [428], it can prevent biofouling without being released into the water. Researchers discovered a correlation between the pungency level of capsaicin analogs and their anti-fouling properties, which was determined by their chemical structures [412,429]. Three derivatives of capsaicin were also created by Yan et al., [430]. All three capsaicin derivatives showed good anti-fouling performance throughout the 186-day evaluation of the anti-fouling capacities in the Qingdao Sea by the authors. However, a study found that the anti-fouling performance diminished after a long (over four months) test in a marine setting [431]. It is challenging to sustain capsaicin's anti-fouling efficiency.

Synthetic analogs are more appealing than the ongoing hunt for novel chemicals in nature because of their effectiveness and affordability. Additionally, analogs' structures may be changed to improve their anti-fouling properties. This does not imply that the search for novel natural chemicals is not essential; however, these molecules may aid in the creation of novel analogs. Materials employed in the synthesis should be ecologically benign and have a high yield based on the criteria of sustainable development. Further research could be done to determine if these analogs might be hazardous to marine creatures that are not their intended targets [420].

5. Conclusion and outlook

Effective and ecologically acceptable anti-fouling techniques are critically required for navigation, detection, engineering, and aquaculture. The creation of nontoxic anti-fouling coatings may benefit from fresh insights gained from nature, which also offers workable solutions to the problem of marine biofouling. Nature-inspired coatings can raise studies in the marine anti-fouling industry. This review discussed several current developments in nature-inspired anti-fouling techniques that imitate the parts, traits, and topographies of living things. The fundamentals, along with the benefits and drawbacks of nature-inspired SLIS, cell membrane-inspired anti-fouling surface, nature-inspired hydrogel, natural products, and nature inspired textures, are all well shown.

- SLIS inhibits surface adhesion; therefore, this strategy may prevent marine biofouling. SLIS lubricants, such as fluorocarbons and silicone oil led to their usage in marine sectors as anti-fouling agents. The low kinematic viscosity of silicone oil makes it easier to lose. Due to the presence of fluorine atoms, fluorocarbon oil has low surface tension, which makes it ecologically unfavorable. Lubricants also may be lost rapidly in various ways, such as evaporating or being taken away when they come into contact with external chemicals. The SLIS could resolve this problem with a self-replenishing lubricant. However, the lubricant's performance may be negatively effect through the progressive and long-term enrichment of foulants. In order to entirely eliminate the issue of failure brought on by lubricant loss, super-slippery surfaces start to utilise lubrication-free slipper materials. Additionally, contamination of adjacent surfaces are prevented since solid lubricants are not as fluid. However, being a new tactic, this subject requires further research.
- Zwitterionic coatings are suitable for anti-fouling but have various drawbacks, including low durability and mechanical strength. One of the long-lasting and most effective anti-fouling solution is PDA-

zwitterionic polymer conjugation. Dopamine grafting with zwitterion to generate homogeneous layers and high zwitterion content on the surface for maximal anti-fouling. PDA-zwitterion conjugation has made significant progress, but it still needs to better understand PDA-zwitterionic polymer formation principles and improve its mechanical resilience. With more profound knowledge of PDA-zwitterionic and associated polymerization processes, new rationally designed materials with high functionality may be created to meet the long-term marine needs for superior anti-fouling materials.

- The inadequate mechanical capabilities of nature-inspired hydrogels still need to be improved. Double network hydrogel (DN) is one of the most intriguing hydrogels available today because it may display enhanced mechanical strength and anti-fouling properties for a long time in the marine field.
- Nature-inspired textures are more ecologically benign because they leverage intensely confined near-surface effects to generate anti-biofouling without releasing hazardous chemicals. The nanostructured surfaces can potentially prevent bacterial infections by either killing the bacteria attached to them or delaying their adhesion and subsequent biofouling development. Mechano-bactericidal nanostructures' physicochemical interactions with bacteria make them superior to chemical antimicrobials. However, nanostructures connected to a substrate surface have not been demonstrated to endure the tear and wear commonly encountered by a massive amount of mimic surface, so the nanostructures may "shed" over time and lose mechano-bactericidal efficacy. Therefore, micro-nanopatterning surfaces modified with biocidal chemical may improve the long-term robust antimicrobial property through a combination of bactericidal and anti-biofouling capabilities, can lead to multifunctionality in the marine field.
- Extracted biological macromolecule may be a good substitute for dangerous anti-biofouling chemicals in light of the new discoveries of synthetic anti-fouling compounds derived from nature that are more bio-friendly. They may be used efficiently for anti-fouling purposes by including natural elements in the correct paint. It could be challenging to construct an anti-fouling layer using natural products, given how fast they degrade in the environment. In that case, there could be more opportunities for natural product-based anti-fouling systems. The combination of biodegradable polymer (polycaprolactone and lactic acid) with more environmentally friendly EBM may be a future development trend. A sustainable manufacturing process for biodegradable polymer with synthetic anti-fouling agents must be developed natural product can necessitates for further investigation for a long time in the marine field.

Developing long-lasting and ecologically benign biofouling control coatings or technologies still requires a lot of effort. The creation of synergetic nature-inspired coatings with biofouling control qualities derived from various approaches can be the focus of future research. While planning for the future, it is essential to continually keep in mind several elements, such as cost-effectiveness, large-scale production, and adaptability to various maritime environments. We anticipate that the thoroughness of this review and its clear emphasis on marine applications will inspire engineers and scientists working in the fields of chemistry, physics, materials science, and other multidisciplinary research areas to develop effective nature-inspired biofouling control technologies and coating for the future of the maritime industry.

CRediT authorship contribution statement

Abid Ali: Writing – original draft, Conceptualization. **David Culliton:** Formal analysis. **Shah Fahad:** Investigation. **Zafar Ali:** Resources. **En-Tang Kang:** Writing – review & editing, Supervision. **Liqun Xu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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